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Experimental and Computational Investigation of a Small-Scale Turbopump Featuring a Barske Impeller and a Supersonic Turbine

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Abstract

Your abstract goes here. Summarise the key motivations, methodology, and findings of your turbopump FYP.

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Nomenclature

a	Speed of sound (m/s)
A, A_{th}	Area; geometric nozzle throat area (m^2)
b	Turbine rotor blade height (mm)
b_1, b_2	Pump blade axial width at stations 1, 2 (mm)
b_c, h_c	Annular casing width, radial depth (mm)
c	Absolute velocity (m/s)
c_p	Specific heat at constant pressure (J/(kg K))
c_s	Spouting velocity (m/s)
C_H, h_0	Lock/Busemann head-correction constants (−)
$d_0 \ldots d_4$	Pump station diameters (mm)
d_m	Turbine mean (pitch) diameter (mm)
E	Energy (J); Young's modulus in eq. (55) (Pa)
f	Friction coefficient (−)
F_R	Radial thrust (N)
g	Gravitational acceleration (m/s^2)
H	Pump head (m)
H_{T2}	Lock's total impeller generated head (m)
H_{NL}	Nozzle loss head (m)
H_3	head at diffuser throat (m)
h	Specific enthalpy (J/kg); Δh_s isentropic drop
I	Second moment of area (m^4)
J	Mass moment of inertia ($kg\,m^2$)
k_d	Barske diffuser kinetic-recovery coefficient (−)
k_N, k_R	Nozzle, rotor velocity coefficients (Weiss) (−)
K_1	Lock's pre-rotation factor (−)
K_v	Valve flow coefficient (eq. (56))
M	Mach number; M_{w3} rotor-inlet relative Mach (−)
$\dot{m}$	Mass flow rate (kg/s)
n	Shaft speed (rpm)
n_q	Specific speed $n\sqrt{Q}/H^{0.75}$ (rpm, m^3/s, m)
NPSH_a, NPSH_3	Available NPSH; NPSH at 3 % head drop (m)
p	Static pressure (Pa); p_{01}, T_{01} total state at station 1; p_v vapour pressure
Δp	Pump total pressure rise (Pa); Δp_{st} static component
P	Power (W); P_{RR} disc friction/churning; P_m mechanical loss; P_w ventilation (windage) power
Pr	Prandtl number (−)
q	Dynamic pressure $\frac{1}{2}\rho c^2$ (Pa)
Q	Volumetric flow rate (m^3/s); Q_L leakage flow; Q_{opt} optimum-flow point
r	Radius (m); recovery factor in eq. (51) only
R	Specific gas constant (J/(kg K))
Re_u	Peripheral Reynolds number $u_2 r_2/\nu_l$ (−)
s_{ax}	Impeller–casing axial clearance (mm)
SG	Specific gravity (−)
T	Temperature (K)
u	Blade (peripheral) speed (m/s)
w	Relative velocity (m/s)
Y	Specific work (J/kg)
z	Blade count (−)
α	Absolute flow angle from circumferential (°)
β	Relative flow / blade angle from circumferential (°)
γ	Heat capacity ratio (−)
ζ	Degree of (partial) admission (−)
ζ_D	Diffuser-nozzle loss coefficient (−)
η	Efficiency: pump overall $\rho g Q H/(\tau\omega)$; η_h hydraulic; η_{ts} turbine total-to-static
μ	Dynamic viscosity (always with a state subscript); Mach angle within the MoC set only
ν	Prandtl–Meyer angle (°); kinematic viscosity is written μ/ρ or ν_l; Poisson's ratio in eq. (52)
ρ	Density (kg/m^3); ρ_b blade material density
σ	Stress, with subscript (eq. (52)); slip factor (Wiesner) in §2; standard deviation in the uncertainty analysis
τ	Shaft torque (N m); τ_{pc} polycarbonate shear strength in eq. (64)
ϕ	Pump flow coefficient $Q/(\pi d_2 b_2 u_2)$ (−)
ψ	Pump head coefficient $2gH/u_2^2$ (−)
ω	Shaft angular velocity (rad/s); $\dot{\omega}$ angular acceleration; ω_s nondimensional specific speed

Structural and instrumentation symbols (scoped to their sections)

B_r	Seal balance ratio (−)
M_b, y	Blade root bending moment (N m); extreme fibre distance (m)
V, V_{ref}, V_{exc}	Voltage; ADC reference and transducer excitation (V)
N	Sample count in oversampling, eq. (57) (−)
$\mathcal{V}$	Stored-gas / vessel volume (m^3)
s_p	Shear-plug perimeter, eq. (64) (m)

Subscripts

$0 \ldots 4$	Station index (machine-scoped, see below)
th, st	Theoretical; static
RR, m, h, ts	Disc friction (churning); mechanical / mean; hydraulic; total-to-static
opt, max	Optimum-flow point; maximum
aw, s, v, L	Adiabatic wall; isentropic; vapour; leakage

Superscripts and operators

$\dot{\square}$, $\bar{\square}$	Time rate; mean / non-dimensionalised
$\square^*$	Critical (sonic) reference / non-dimensional (MoC set)
$\square^e$	Effective value, e.g. c^e_{3u}

Stations (machine-scoped)

Pump	0 suction/eye inlet · 1 blade leading edge · 2 blade tip · 3 diffuser throat · 4 discharge
Turbine	1 stator inlet plenum · 2 nozzle throat · 3 nozzle exit/rotor inlet · 4 rotor exit

Scoped symbol sets (declared here, used only in their subsections)

MoC set (section 2.5.1)	$M^* = c/a^*$, $a^* = \sqrt{\gamma R T^*}$, r^*, R^*, x^*, y^*, μ_k (Mach angle), m_k, $\Delta\nu$
Starting set (section 2.5.1)	M^*_l, M^*_u surface critical Machs; K^*_{max} weight-flow constant; Q vortex parameter; C 2I
Boundary layer set (Sasman–Cresci)	H_i shape factor, θ momentum thickness, f, C_f, g_w, T_e, $\bar{T}$, s/c, δ^* displacement thickn

Abbreviations

ADC	Analogue to Digital Converter
BEP	Best Efficiency Point
CFD	Computational Fluid Dynamics
DAQ	Data Acquisition
ESC	Electronic Speed Controller
FEA	Finite Element Analysis
GUI	Graphical User Interface
MoC	Method of Characteristics
P&ID	Piping and Instrumentation Diagram
SLA	Stereolithography (resin printing)

1 Introduction

Rocket engines require high pressures to operate efficiently. Small scale liquid propulsion such as satellite propulsion and amateur liquid roket propulsion runs almost entirely on pressure fed feed systems due to their simplicity and reliability. Larger orbital rockets use turbopumps because the mass of the required tanks becomes prohibative. However, as space launch becomes exponentially more accessible, there is a growing interest in small scale rocket propulsion systems. For example, lunar landers, satellite propulsion and amateur liquid rockets are seeing a resurgence in interest. Due to the historical focus on large turbopumps for such applications, turbopumps at the smaller scale are not as well documented and have different design considerations. A turbopump is a pump and the turbine that drives it, and both are punished by the small scale.

1.1 Small scale rocket pump challenges

Pump driven propulsion is the solution orbital liquid rockets use to solve the tank mass problem. By storing propellents at lower pressures and pumping them to the required pressures allows for a large reduction in tank mass. The pump is usually driven by turbines powered by burning a portion of the propellents, or by the heat generated by the cooling of the engine. In recent years, battery and electric pump technology has advanced to such a point that for smaller rockets such as the Electron rocket by Rocket Lab [?], electric pumps are starting to become a more viable alternative to turbine driven pumps.

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Pumps are generally characterized by their flow rate Q, the head H and rotor speed n [?]. These three parameters largely decide the pump geometry and type of pump. These three parameters can be related by a quantity known as specific speed, n_q, N_s or ω_s depending on the units and definitions used:

$$n_q = \frac{n\sqrt{Q}}{H^{3/4}} \quad [\mathrm{RPM}, \mathrm{m^3/s}, \mathrm{m}] \tag{1}$$

Different pump geometries are optimal for different regions of specific speed as can be shown by fig. 1. The radial centrifugal impeller reaches its best efficiency roughly between an n_q of 10 and 40 [4, ?], and rocket engine pumps are normally sized to sit in this band, splitting the head across multiple stages when a single stage would otherwise fall below it. This works great for large rocket engines, but scaling this down to smaller engines becomes increasingly difficult. All rocket engines require high pressures, but smaller rocket engines require less flow rate. To stay in the specific speed according to eq. (1) that allows for high efficiency in the centrifugal pump requires increasing rotational speed.

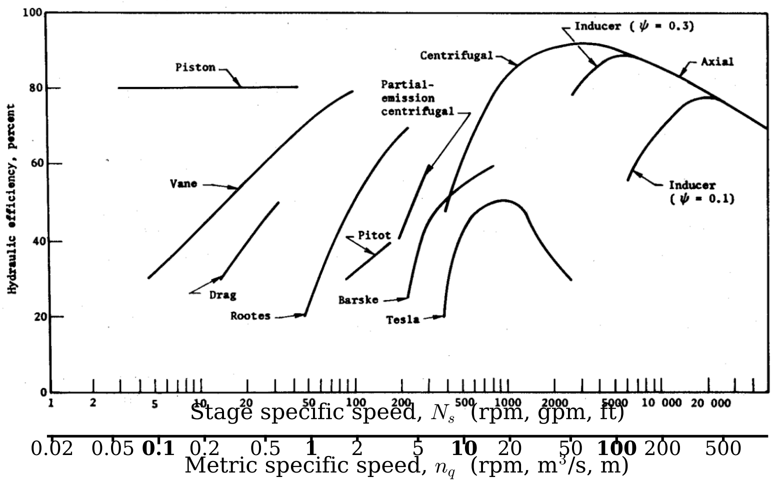

Figure 1: Optimum pump geometry and attainable efficiency against specific speed, with the design point of this thesis at $n_q = 6.4$ marked. Most single-stage rocket engine pumps sit in the radial impeller optimum of n_q 10 to 40.

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To achieve the specific speed n_q of 20 for a smaller rocket engine of around 5 kN requiring 1.0 kg/s of kerosene at 50 bar, eq. (1) leads to rotational speeds of around 71 051 rpm. Such high rotational speeds leads to challenging mechanical design conditions. To solve this problem, a single pump can be separated into multiple stages, separating the head rise to increase the specific speed of each stage. This is a common solution to pumping hydrogen. This thesis however addresses the other solution, to study a single stage pump geometry at this low specific speed. In rocket applications, the lower component efficiency could trade better by other factors such as lower mass, cost, reliability and simplicity. As shown in fig. 1, at lower specific speeds, the partial emission pump, of which the Barkse impeller is the main form, becomes the optimum efficiency pump geometry at specific speeds of 5-10. The term partial emission refers how the

the pump outlet only taps a small arc of the pump periphery. Full emission pumps are designed to have the impeller discharge around the whole annulus, collecting flow through a continuous volute. Literature uses the names partial emission and Barske interchangeably.

However, even though these pumps are the most efficient at these low specific speeds, they physically cannot achieve the high efficiencies of a full emission pump. This is the main limitation of trying to operate at low specific speeds. Due to its lower efficiency and the lack of pump fed liquid rockets for small engines, there is a lack of literature, design knowledge and experimental data surrounding these pumps. This thesis aims to address this gap by reviewing existing literature, designing and manufacturing a partial emission centrifugal pump and comparing its experimental performance specifically in the use case of rocket engines.

The lack of data is specifically a lack of experimental data. Open studies of this regime do exist, but they are mostly numerical. A Barske type impeller at $n_q = 4.7$ was studied with CFD and reached an efficiency of 31.5% [5]. A conventional closed impeller was designed at the same specific speed as this thesis, but it was CFD only, unvalidated, and required closed channels of 0.31 mm width [6]. A partial emission rocket e-pump for a lunar lander was designed for 630 m at 40,000 RPM, again in CFD [7]. What is scarce is open experimental characterisation at this specific speed, with a measured loss breakdown and measured cavitation behaviour, and that is what this work supplies. A NASA trade study of low thrust rocket pumps covers the same trade and carries the Barske pump as a candidate [8].

1.2 Small scale rocket turbine challenges

On the other end of the turbopump shaft is the turbine which drives the pump. The turbine faces the same problem as the pump. If using the gas generator cycle, the small turbine leads to lower blade speeds u. To extract the maximum work from from the gas, gas velocity c_s must be high, leading to a supersonic blade turbine. The high gas velocity relative to blade speed leads to the turbine running at a lower efficiency than the optimum $u/c_s \approx 0.5$ [**?**]. The smaller mass flow also cannot fill the annulus at a sensible blade height, leading to partial admission where only a fraction of the annulus has gas flowing through it at any given time. This makes the stage lossy and awkward to design. In addition to these design limitations, supersonic turbine design is far less studied than subsonic, where methods are scattered across older sources.

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This thesis tackles the design of a turbopump in the context of small scale liquid rocket engines which forces both pump and turbine off optimum design conditions, then demonstrates their coupled performance in a full integration test. The thesis is framed around the design of a 20 000 rpm turbopump using water as a inert simulant for the pumped rocket propellents. Then, a supersonic partial admission impulse turbine is designed to drive the pump using compressed air as the working fluid. The two are then coupled together and tested as a single system to evaluate performance.

1.3 Scope and Limitations

This thesis covers the theory, design, manufacture and testing of a small scale turbopump system on inert propellents. The pump and turbine were tested up to 9000 rpm, significantly lower than the design speed of 20 000 rpm in the first and final iterations of the conducted experiments. This means the data validation can only happen for the off-design point and extrapolation to the design point is required. Cryogenic and hot gas design and testing are strictly outside the scope of tis thesis. Operational safety procedures are included in the appendix.

1.4 Aims and Objectives

The aim of this thesis is to investigate the performance of a small scale rocket low specific speed turbopump with a Barske pump and a supersonic partial admission turbine. To this end, the objectives of this thesis are:

1. Pump: Assemble a Barske pump design methodology from the literature, use it to design and build the pump, then characterise it experimentally for head-flow, efficiency and cavitation, and compare the measurements against the methodology's predictions and prior literature.

2. Turbine: Assemble a design methodology for a single-stage partial-admission supersonic impulse turbine from the literature, use it to design and build the rotor, then validate those models against the achievable test data.

3. Turbopump: Couple the pump and turbine and demonstrate them running as a single machine, then use that result to check whether the combined methodology predicts a working turbopump of this class.

1.5 Thesis Structure

Chapter 2 establishes the theoretical background for both machines: the Barske pump and its sizing and loss models, and the supersonic impulse turbine theory including blade profile design and loss models. Chapter 3 applies these models to design the actual pump and turbine, including the mechanical and structural design. Chapter 4 describes the experimental setup: the test configurations, the water and air circuits, and the custom instrumentation and data acquisition system. Chapter 5 presents the experimental results and discusses them against the design predictions and prior literature. Chapter 6 draws conclusions against the objectives and recommends future work.

2 Literature and Theory

Much of the supersonic blading design methodology dates from the 1950s and 60s [9, 10]. The same analytical core, method of characteristics nozzle and blade passage design remains foundation of current work. Sudhof's 2020 dissertation [11] frames modern developments into design as a paradigm shift. Older methods studied analytical profiles first then checked practical restrictions such as finite leading edge thickness, shock boundary layer interaction afterwards. Modern computational capacity lets the blade be designed and optimised around these restrictions from the start. Griffin and Dorney demonstrated this on a two-stage supersonic turbine, where CFD, response-surface and neural-net optimisation of a conventionally designed baseline gave a measured total-to-static efficiency gain of 9 points [12]. These methods are still the basis of current small turbine design studies [13, 14]. On the pump side, the Barske architecture is seeing renewed interest [5], but literature is still very scarce [7].

2.1 Pump Theory

2.1.1 Euler Turbine Equation

The working principles for pump and turbine design can be derived from the conservation of angular momentum by Newton's second law of motion. In turbomachinery, this is usually expressed in terms of the Euler turbine equation shown in eq. (2), where c is velocity in the inertial frame, w is in the rotatating frame and u is the local circumferential velocity as shown in fig. 2.

$$P = \tau\omega = \rho Q\left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = \rho Q\left(u_2 c_{2u} - u_1 c_{1u}\right) \tag{2}$$

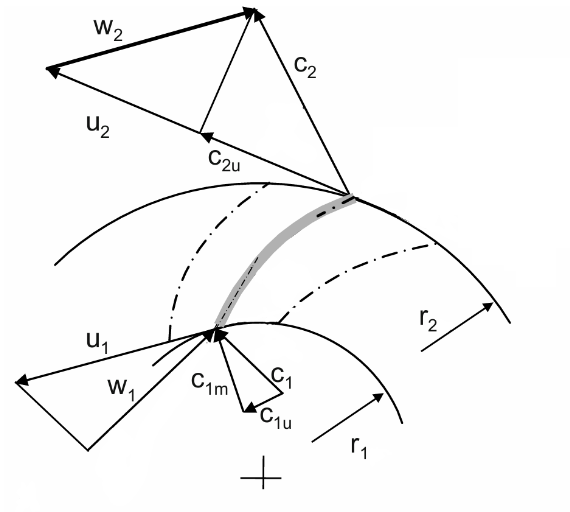


Figure 2: Velocity triangles in a conventional pump, adapted from Gulich

Rewriting this equation in terms of specific work $Y = P/\rho Q$ per unit mass:

$$Y = \left(r_2 c_{2u} - r_1 c_{1u}\right)\omega = u_2 c_{2u} - u_1 c_{1u} \tag{3}$$

derive from bernouli, with rotating frame substitution $c^2 = w^2 - u^2 + 2 \times u \times c_u$ and neglecting compressibility losses:

$$p_1 + \frac{\rho}{2}w_1^2 - \frac{\rho}{2}u_1^2 = p_2 + \frac{\rho}{2}w_2^2 - \frac{\rho}{2}u_2^2$$

and as such the theoretical rise in static pressure:

$$\Delta p_{st} = p_2 - p_1 = \frac{\rho}{2}\left(w_1^2 - w_2^2\right) + \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{4}$$

2.1.2 Non-dimensional Performance and Scaling

Understanding why low specific speed pumps are less efficient, and the design choices that follow from it, comes down to a few non-dimensional parameters so they are defined first. The following derivation uses velocity terms shown visually in fig. 2. A pump adds energy to the fluid by spinning it up towards the speed of the impeller rim, u_2. The work it does per unit mass is the rim speed times the swirl velocity it adds to the fluid, $u_2 c_{2u}$, and that swirl is proportional to the rim speed, so the head a pump can generate scales with u_2^2. The flow rate meanwhile leaves the impeller radially through the exit area $\pi d_2 b_2$ at the meridional velocity c_{m2} where b_2 is the blade width at the exit, so $Q = \pi d_2 b_2\, c_{m2}$. Measuring the head against the rim-speed head u_2^2, and the radial velocity against the rim speed u_2, gives the head and flow coefficients, following Gülich [4]:

$$\psi = \frac{2gH}{u_2^2}, \qquad \phi = \frac{c_{m2}}{u_2} = \frac{Q}{\pi d_2 b_2 u_2} \tag{5}$$

So ψ is the fraction of the achievable rim-speed head that the machine actually delivers, and ϕ is how fast the fluid moves out radially compared to how fast the rim turns. Between them ψ and ϕ set the tangential and radial sides of the exit velocity triangle relative to u_2, so two operating points at the same ϕ have the same triangle shape and the same flow physics whatever the size or the speed.

2.2 Main Loss Mechanisms at Low Specific Speed

Losses in pumps can be separated into power losses and head losses. Power losses increase the shaft power required to drive the pump but does not directly affect the head delivered, while head losses directly reduce the head from the ideal value.

2.2.1 Disc Friction Loss

When a rotor spins in a fluid, the shear stress in the fluid acts on the rotor faces as a drag torque. In most centrifugal pumps the impeller is broadly disc shaped, so this is called disc friction loss. Taking the main parameters of the equation from Gulich [4], the power lost to disc friction scales as

$$P_{RR} \approx k_{RR}\, \rho\, \omega^3 r_2^5 \tag{6}$$

where k_{RR} is a friction coefficient that falls slowly with Reynolds number. By dividing this by the useful power $P_{hyd} = \rho g Q H$ and substituting the head coefficient $H = \psi u_2^2/2g$ with $u_2 = \omega r_2$ gives:

$$\frac{P_{RR}}{P_{hyd}} \approx \frac{k_{RR}\, \rho\, \omega^3 r_2^5}{\rho g Q H} \approx \frac{k_{RR}\, \omega^3 r_2^5}{gQ} \cdot \frac{2g}{\psi\, \omega^2 r_2^2} = \frac{2k_{RR}\, \omega r_2^3}{\psi\, Q} \tag{7}$$

Then, by substituting the head coefficient into specific speed and rearranging for Q:

$$n_q = \frac{n\sqrt{Q}}{H^{3/4}} \propto \frac{\omega\sqrt{Q}}{(\psi\, \omega^2 r_2^2)^{3/4}} = \frac{\sqrt{Q}}{\psi^{3/4}\, \omega^{1/2} r_2^{3/2}} \Rightarrow \quad n_q^2 \propto \frac{Q}{\psi^{3/2}\, \omega\, r_2^3} \qquad \Longrightarrow \qquad Q \propto n_q^2\, \psi^{3/2}\, \omega\, r_2^3 \tag{8}$$

Back into the ratio:

$$\frac{P_{RR}}{P_{hyd}} \approx \frac{2k_{RR}\, \omega r_2^3}{\psi\, Q} \propto \frac{k_{RR}\, \omega r_2^3}{\psi \cdot n_q^2\, \psi^{3/2}\, \omega\, r_2^3} = \boxed{\frac{k_{RR}}{n_q^2\, \psi^{5/2}}} \tag{9}$$

This is the reason low specific speed pumps cannot reach high efficiency. The relative disc friction scales as $1/n_q^2$, so halving the specific speed quadruples the share of the input power lost to the disc. As an example point on the size of this, Gülich gives the disc friction as roughly 30% of the useful power of the impeller at $n_q = 10$, $\psi \approx 1$ and $\mathrm{Re} = 10^7$ [4]. Gulich provides fig. 3 to show this relationship visually.

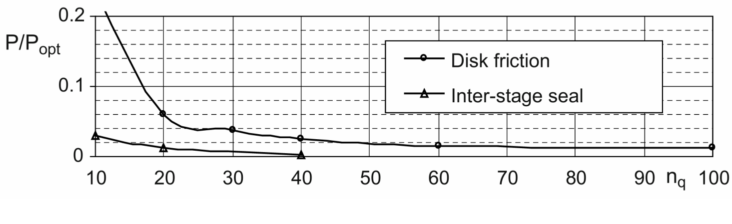


Figure 3: Disc friction loss vs specific speed n_q after Gülich. At low specific speed the disc friction dominates.

The head coefficient in the denominator of eq. (9) also points the pump designer to design a pump with a higher head coefficient as $P_{RR} \propto \psi^{-5/2}$. For a fixed desired head H and rotational speed n, a high head coefficient means a smaller tip speed and thus a smaller diameter can be derived from **??**:

$$u_2 = \sqrt{\frac{2gH}{\psi}} = \frac{\pi d_2 n}{60} \tag{10}$$

Since $P_{RR} \sim r_2^5$ shrinking the diameter collapses the disc friction. This is why Gülich recommends maximising the head coefficient at low specific speed.

2.2.2 Hydraulic Losses

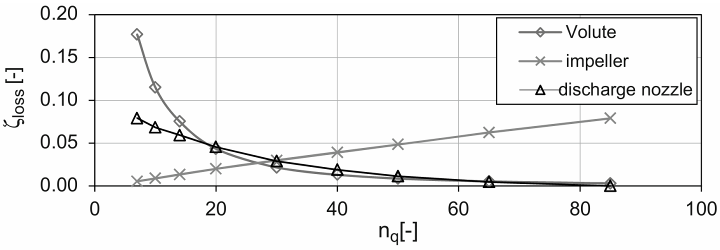
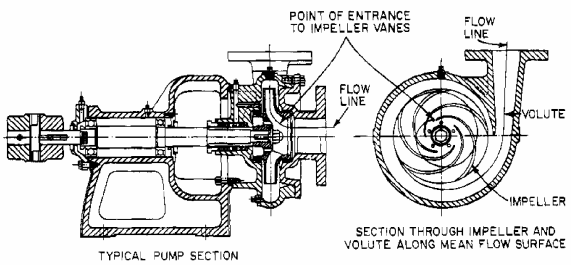


Figure 4: Breakdown of pump losses against specific speed, after Gülich. At low specific speed the disc friction dominates while the impeller (blade) losses are small.

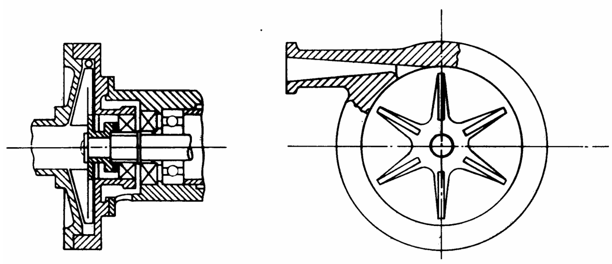
fig. 4 shows the hydraulic loss breakdown against specific speed. At low specific speed the impeller blade losses are small, because the velocities of the flow relative to the blades are low, while the volute and the discharge nozzle losses become the next largest term.

2.3 The Barske Partial-Emission Pump

The Barske pump, also known as the partial-emission pump [?], forced vortex pump [15] is a type of centrifugal pump found to be effective in these conditions. The Barske Pump is very simple in geometry as shown in fig. 5b as compared to a conventional pump such as fig. 5a. The impeller consists of a flat disc with straight radial blades. The volute is swapped for cylindrical (annular) casing, with a divergent diffuser placed tangent to the casing. Because the fluid leaves the impeller at high velocity relative to both the casing and the diffusor, the surface finish becomes a primary concern for efficiency at this scale [?]. Gülich also notes that for $n_q < 12$ an annular casing can reach higher efficiencies than a volute, the spiral collector of a conventional pump [?], which has the side benefit of being far simpler to manufacture, a real advantage at small scale where tight tolerances are hard to hold. Conversely, as the losses in the impeller are small, the Barske impeller surface finish is not critical and can be left rough.



(a) Conventional shrouded full emission impeller



(b) Barske open partial emission impeller

Figure 5: Comparison of the conventional and Barske pump architectures. The Barske impeller is open, has straight radial blades, and discharges into an annular casing with a single tangential diffuser.

The Barske impeller pump design which was developed for the need for manufacturing simplicity and low specific speed applications [16] became one of the main centrifugal pump designs at this specific speed. The Barske impeller operates differently to a normal full emission pump. In a conventional centrifugal pump, impeller blade geometry aims to perfectly guide fluid flow along the blade surfaces in a smooth spiral streamline out of the impeller into the volute [4]. The Barske impeller is instead designed to induce almost a pure forced vortex flow where $u = \omega r$ and the fluid is essentially rotating as a solid body with the impeller [16] where meridonal or radially outward flow is negligable [17]. This also explains the unimportance of surface finish on the blades, as low relative velocity between the fluid and the impeller means lower friction losses. Another difference between conventional impellers and the barske impeller is the absence of shrouds. The open impeller choice is to further reduce disc friction losses as in the case of radial impellers, shroud friction far exceeds contributions the contribution of all other components [4]. Conventional impellers uses shrouds to guide the flow and reduce secondary flows, but as the Barske impeller operates on a wholely different principle, this shroud is not needed. Removing the shrouds also reduces axial thrust loads on the impeller as there is less surface area for the fluid to push against, which assists with sealing and mechanical design of the pump shaft.

2.3.1 Barske Pump Performance

To predict the performance of a Barske pump, Barske starts by assuming that the flow is nearly a pure forced vortex as shown in fig. 6. He also assumes the relative velocity w is assumed to be negligable. This turns eq. (4) into eq. (11), which is the theoretical static pressure rise of a Barske impeller.

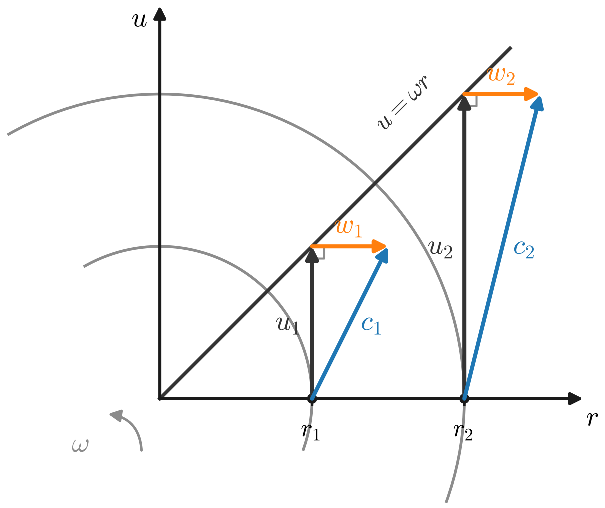


Figure 6: Velocity triangles in a forced vortex pump

$$\Delta p_{st} = \frac{\rho}{2}\left(u_2^2 - u_1^2\right) \tag{11}$$

The kinetic energy increase assuming negligible inlet prerotation is:

$$q = \frac{\rho}{2}u_2^2 \tag{12}$$

eq. (12) is the theoretical dynamic pressure a barske impeller generates at the blade tips. In a barske pump, the diffuser is responsible for converting this kinetic energy into static pressure by slowing down the fluid in an expanding diffuser. Barske introduces a loss coefficient for the kinetic energy which can be recovered through the diffuser to be between 0.2 and 0.5. Barske uses the symbol ψ for this coefficient, but as it conflicts with the commonly used symbol for head coefficient, this paper will instead use k_d. Combining the static and kinetic pressure rise, the total increase in pressure is:

$$\Delta p = \Delta p_{st} + k_d\, q = \frac{1}{2}\rho\left((1+k_d)u_2^2 - u_1^2\right) \tag{13}$$

This simple model is used to generate the dimensions in section 2.3.2, but a more refined model from Lock [15] is described in section 2.3.4 to more accurately predict the full characteristics of the pump at any operating point.

2.3.2 Barske Sizing Method

The barske pump is sized by the methods provided by barske [17] with the assitance of Gulich. First the inlet is sized by choosing a reasonable inlet velocity and calculating the inlet area:

$$d_0 = \sqrt{\frac{4\dot{m}}{\pi\rho c_0}} \tag{14}$$

The impeller blade inlet diameter is enlarged by a small amount as some pre-rotation is desirable for increased head. For example, $d_1 \sim 1.1d_0$.

By substituting $u_2 = \pi n d_2/60$ into eq. (13) and solving for d_2, the required impeller tip diameter can be calculated from mass continuity:

$$d_2 = \sqrt{\frac{1}{1+k_d}\left(\frac{7200\Delta p}{\rho n^2\pi^2} + d_1^2\right)} \tag{15}$$

Barske then gives these guidelines for other dimensions of the pump:

$$s_{ax} \leq \min\left(0.75, \frac{d_2}{100}\right) \qquad b_1 \geq \frac{1}{4}d_1 \qquad b_2 \geq \max\left(b_1\frac{d_1}{d_2}, 3s_{ax}\right) \tag{16}$$

For the annular casing sizing, Gulich's guideline where B/H = 1.5 to 2.5 can be followed:

$$B = 2s_{ax} + b_2 \qquad H = B/2 \tag{17}$$

To size the diffuser, Barske suggests sizing the diffuser throat to be around 0.8 to 0.9 of u_2, with the area of the diffuser outlet to be 3-4 times the area of the diffuser throat.

2.3.3 Efficiency and Disc Friction

Barske assumes that the disc friction losses can be approximated as a circular disc in a rotating fluid. Converting the equation into metric units gives:

$$P_{RR}[\mathrm{W}] = 1956 \times \rho\nu^{0.2}\left(\frac{n}{1000}\right)^{2.8}\left(d_2^{4.6} + 4.6d_1^{3.6}b_1\right) \tag{18}$$

As the hydraulic efficiency is defined as the useful power out of the pump over the total power into the pump, efficiency can be defined as:

$$\eta_h = \frac{\text{Useful power}}{\text{Input power}} = \frac{P}{P_{th} + P_{RR}} = \frac{Q\Delta p}{Q\Delta p_{th} + P_{RR}} = \frac{\Delta p}{\Delta p_{th} + P_{RR}/Q} \tag{19}$$

If total shaft efficiency is desired, the mechanical losses P_m can be included:

$$\eta = \frac{\Delta p}{\Delta p_{th} + (P_{RR} + P_m)/Q} \tag{20}$$

2.3.4 Lock's Refined Model

Lock provides a refined model which takes into account prerotation, nozzle losses and correction coefficients taking into account theoretical two dimensional flow solutions by A. Busemann and Wislicenus [15, 18]. Lock uses busemann's corrections and derives a refined total head generated by the impeller H_{T2} as eq. (21), where C_H and h_0 are the Buseman correction factors taken from his figure 10A and 10B respectively. K_1 is the prerotation factor which he says himself is arbitrary, and Q_{opt} is the optimum flow rate at which the pump operates at maximum efficiency.

$$H_{T2} = h_0\frac{u_2^2}{g} - C_H\,K_1\left(1 - \frac{Q}{Q_{opt}}\right)\frac{d_1}{d_2}\,\frac{u_2^2}{g} \tag{21}$$

He then uses a diffuser loss coefficient $\zeta_D = 0.194$ to account for diffuser lossess H_{NL} as shown in eq. (22). This coefficient was taken from the Hydraulic Institute's Pipe Friction Manual [19].

$$H_{NL} = \zeta_D\left[1 - \left(\frac{d_3}{d_4}\right)^2\right]^2\frac{c_3^2}{2g} \tag{22}$$

Finally, to find the static head developed across the pump, the total head generated by the diffuser H_{T2} is reduced by the diffuser losses H_{NL}, dynamic head at the diffuser exit $c_4^2/2g$ and the dynamic head at the impeller inlet $c_0^2/2g$ as shown in eq. (23).

$$H = H_{T2} - H_{NL} - \frac{c_4^2}{2g} + \frac{c_0^2}{2g} \tag{23}$$

Equation eq. (21) requires the parameter Q_{opt}, which is solved by expressing eq. (23) as a function of flow rate Q and taking the derivative with respect to Q and setting it to zero. By expressing $c^2/2g$ by $(8/g\pi^2)Q^2/d^4$ with $c = 4Q/\pi d^2$, the head can be expressed as a function of flow rate Q as shown in eq. (24).

$$H = h_0\frac{u_2^2}{g} - C_H\,K_1\left(1 - \frac{Q}{Q_{opt}}\right)\frac{d_1}{d_2}\,\frac{u_2^2}{g} - \frac{8Q^2}{g\pi^2}\left(\frac{\zeta_D}{d_3^4} + \frac{1}{d_4^4} - \frac{1}{d_0^4}\right) \tag{24}$$

Solving the derivative for Q_{opt} gives the optimum flow rate as shown in eq. (25).

$$Q_{opt} = \sqrt{\frac{h_0\;u_2^2}{\frac{24}{\pi^2}\left(\frac{\zeta_D}{d_3^4}\left(1 - \left(\frac{d_3}{d_4}\right)^2\right) + \frac{1}{d_4^4} - \frac{1}{d_0^4}\right)}} \tag{25}$$

Lock and Barske suggest that the flow cutoff point is limited by cavitation at the diffuser throat. To find the static pressure at the diffuser throat, reversing the transfer between dynamic head into static head provided by the previous equations:

$$H_3 = H + \frac{8Q^2}{g\pi^2}\left(\frac{\zeta_D}{d_3^4}\left(1-\left(\frac{d_3}{d_4}\right)^2\right)+\frac{1}{d_4^4}-\frac{1}{d_3^4}\right) \tag{26}$$

Then solving for the Q at which H_3 is equal to vapor pressure head $H_{vap} = \frac{1}{\rho g}(p_{vap} - p_{inlet})$ gives the flow cutoff point.
An observation by Lock should be kept in mind when his model is compared to data later. He notes that rounding of the peripheral leading edges reduces the effective head [15]. This is evidently not captured in the model and could one of the many reasons for discrepancies between theory and experiment.

2.4 Turbine Theory

2.4.1 Impulse Turbines

Turbines work with the same Euler turbine equation. In the supersonic impulse turbine design, the blades are only occupy a small outer portion of the blisk ($d_3 \approx d_4 \approx d_{\text{mean}}$, $u_3 \approx u_4 \approx u$), so the euler turbine equation eq. (2) can be simplified:

$$\Delta h_{\text{use}} = c_{3u}u_3 - c_{4u}u_4 = u(c_{3u} - c_{4u}) \tag{27}$$

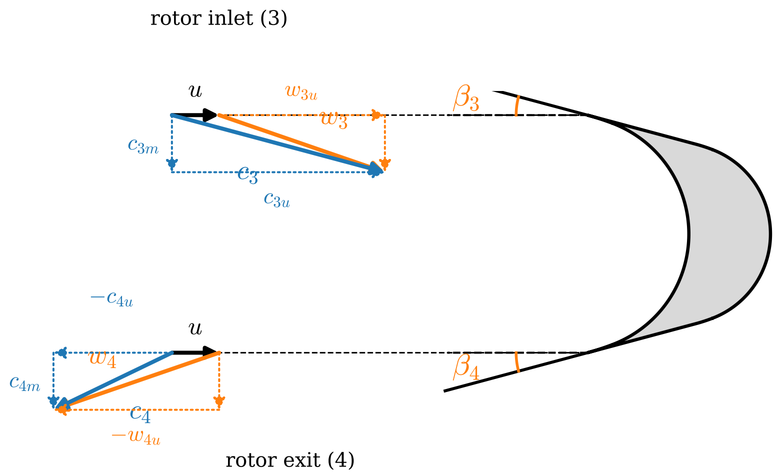


Figure 7: Impulse turbine velocity triangle nomenclature at the rotor inlet (station 3) and exit (station 4). The design point triangles with actual values are shown in fig. 18.

This following derivation follows from Sudhof's [11].From eq. (27), it is understood that a higher flow velocity difference $c_{3u} - c_{4u}$ per unit mass will lead to a higher specific work Δh_{use} according to equation eq. (3). Maximizing $c_{3u} - c_{4u}$ as visualized by fig. 7 is done by maximizing the absolute velocity and as close to 180° of turning as possible. For this reason, gas generator cycle turbines tend to take advantage of the high pressure ratios and thus supersonic velocities. The amount of flow turning is usually limited by geometrical and mechanical design reasons.

2.4.2 Impulse Turbine Efficiency

The ideal efficiency of the impulse stage follows from eq. (27) and the velocity triangles in fig. 7. At the rotor inlet and exit the circumferential components of the absolute and relative velocities are related by

$$c_{3u} = w_{3u} + u, \qquad -w_{4u} = -c_{4u} + u \Rightarrow c_{4u} = u + w_{4u} \tag{28}$$

The ideal impulse blade turns the relative flow without slowing it down, so the exit relative velocity mirrors the inlet one, $w_{4u} = -w_{3u}$. Substituting into eq. (27),

$$\Delta h_{\text{use}} = u\,(c_{3u} - c_{4u}) = u\,[(w_{3u} + u) - (u + w_{4u})] = 2u\,w_{3u} = 2u\,(c_{3u} - u) \tag{29}$$

As ideally, nozzle exit circumferential velocity equals the circumferential velocity at the blade inlet, $c_{2u} = c_{3u}$, eq. (29) gives the required circumferential velocity the nozzle must deliver for a given specific work Δh_{use} and blade speed u. The nozzle exit absolute velocity c_3 is related to the circumferential component by the nozzle angle β_3 measured from the circumferential direction as shown in eq. (30).

$$c_{3u} = \frac{\Delta h_{\text{use}}}{2u} + u, \qquad c_3 = \frac{c_{3u}}{\cos\beta_3}, \qquad c_{3m} = c_3 \sin\beta_3 \tag{30}$$

Dividing the work eq. (29) by the kinetic energy $c_3^2/2$ the nozzle supplies gives the stage efficiency. In a pure impulse stage the entire expansion happens in the nozzle, so the available energy per unit mass is the jet kinetic energy $c_3^2/2$, and with $c_{3u} = c_3 \cos\beta_3$.

$$\eta_{\text{ideal}} = \frac{2u\,(c_3 \cos\beta_3 - u)}{c_3^2/2} = 4\,\frac{u}{c_3}\left(\cos\beta_3 - \frac{u}{c_3}\right) \tag{31}$$

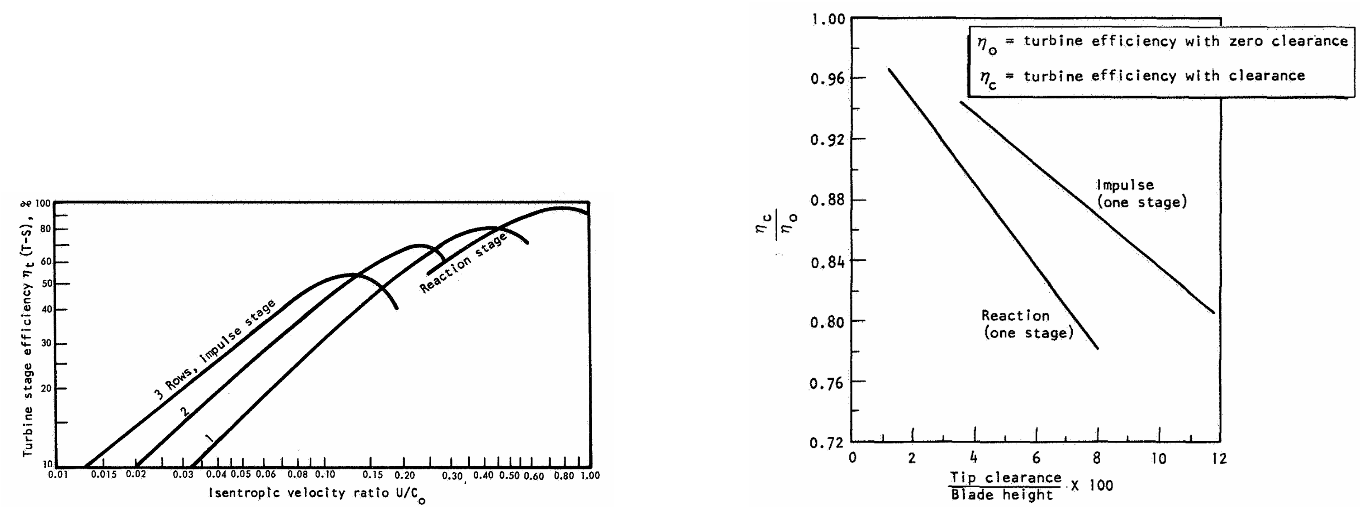
The efficiency depends only on the blade to jet speed ratio u/c_3 and the nozzle angle. By taking the derivative of eq. (31) and setting it to zero, the optimum sits at $u/c_3 = \cos\beta_3/2$. For most turbines with a low β_3 the optimum is around $u/c_3 \approx 0.5$.
The jet spouting velocity c_3 used in this ratio is the spouting velocity, the velocity reached by an ideal isentropic expansion of the ideal gas over the full stage pressure ratio,

$$c_3 = \sqrt{2\,\Delta h_{1\to 3}} = \sqrt{2\,c_p T_{01}\left[1 - \left(\frac{p_3}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{32}$$

For 300 K air with a pressure ratio of 10, $c_3 = 538\,\text{m/s}$. A 100 mm diameter turbine on a 20,000 RPM shaft has a blade speed of around 105 m/s, leading to a u/c_3 of 0.19. The spouting velocity of even a moderate pressure ratio gas generator at a higher outlet temperature can quickly reach up to velocities c_3 from 800 m/s upwards, so u/c_3 is forced far below the optimum of around 0.5. This is a fundamental limitation of small diameter turbines.

2.4.3 Impulse vs Reaction Turbines

NASA published fig. 8a in its SP-8110 showing that for low u/c_3 turbines, impulse turbines are more efficient, even if reaction turbines can have a higher overall efficiency. In addition, the same conclusion is drawn by comparing the influence of tip clearance on the two types, where impulse turbines are more robust to tip clearance as shown by fig. 8b. This is especially important for small scale turbines where manufacturing tolerances are more difficult to achieve and contribute to more significant losses [**?**].



(a) Turbine efficiency vs u/c_3

(b) Effect of tip clearance on turbine efficiency

Figure 8: Comparison of efficiency between impulse and reaction turbines against u/c_3 and tip clearance.

2.4.4 Blade Height and Partial Admission

Another problem with designing small turbines is with the low blade height required. By calculating the mass flux through the turbine, the blade height can be designed:

$$H = \frac{\dot{m}}{\rho_3 c_{3m} d_{\text{mean}}} \tag{33}$$

Where ρ_3 is the density of the turbine gas at the turbine inlet calculated using isentropic gas relations. Using the example turbine above (a 2 kW stage with $\dot{m} = 23.0\,\text{g/s}$ at $\rho_3 = 2.24\,\text{kg/m}^3$), the required full admission blade height would be 0.74 mm, which is far too small to manufacture or to run without the wall boundary layers swallowing the whole passage. Blade height as small as 1.25 mm on a turbine diameter of 50 mm has been reported to work in practice by [20]. This is in line with their internal design rules, where a blade height to mean diameter of h/d_{mean} from 0.025 to 0.10 is recommended. However, NASA's SP-8107 recommeneds a minimum height of 3.8 mm [21]. By introducing partial admission, the mass flow through the turbine can be restricted to a smaller arc around the annulus at a slight cost to efficiency. The partial admission ratio ζ can be included into the blade height formula as so:

$$H = \frac{\dot{m}}{\zeta \rho_3 c_{3m} d_{\text{mean}}} \tag{34}$$

By reducing the admission to $\zeta = 15\,\%$, the same mass flow is forced through a smaller arc of the annulus and the blade height is increased to 4.92 mm. The blade height matters because it sets the blade aspect ratio H/c, the blade height over chord, which fixes how much of the passage the endwall boundary layers block.

Ohlsson found that blade efficiency and nozzle efficiency both drop with low aspect ratio blades. He tested 4 turbines with aspect ratios ranging from 0.07 and 0.7 on a 4 inch diameter turbine. He found that the efficiency of both the nozzle and the rotor dropped linearly to the inverse of the aspect ratio, as shown in eq. (35) [22].

$$\eta_{\text{nozzle}} = 0.92 - 0.016(c_{\text{nozzle}}/H_{\text{nozzle}}) \qquad \eta_{\text{rotor}} = 0.91 - 0.10(c_{\text{rotor}}/H_{\text{rotor}}) \tag{35}$$

This loss of efficiency with low aspect ratio has to be balanced with the loss of efficiency with partial admission. Weiss provides eq. (38) for the power loss having to drag the non-admitted arc through stagnant air as a function of partial admission ratio ζ and blade height H. The other loss present only in partial admission turbines is the loss asssociated with the filling and empting the blade channels. Ohlsson found that having a single continous arc of admission was more efficient than distributing arcs of admission around the annulus as the entering loss is repeated [23]. NASA SP-8110 agrees with this, but also notes that sometimes this continous nozzle arc has to be violated if designs cannot sustain the uneven loads [24].

2.4.5 Loss Model

The main loss model used in the design of the turbine is the one provided by Weiß [**?**]. There are more general loss models provided by texts such as NASA SP-290 [25], giving loss models for tip-clearance, disk friction, partial admission and incidence losses. However, those models were built on large full-admission, subsonic to transonic turbines. Weiss specifically derived his loss model for the specialized application of a supersonic impulse turbines at small scales (1-200kW) [20]. As the regime of the turbine considered in this thesis matches closer to Weiss's model compared to SP-290's model, Weiss's model is used. Weiss describes different models lead to stage efficiency differences of about 10% points. Derived from rocket turbopumps literature and adjusted by in-house measurements, Weiss provides three empirical relations to form his loss model. Weiss references Traupel [26], Craig [27] and Kacker's [3] models for these larger subsonic full admission turbines.

$$k_N = \sqrt{1 - (0.0029\, M_3^3 - 0.0502\, M_3^2 + 0.2241\, M_3 - 0.0877)} \tag{36}$$

eq. (36) is the nozzle loss coefficient, relating the actual velocity to the theoretical isentropic velocity by $k_N = c_{No}/c_{No,is}$.

$$\begin{aligned} k_R = 0.957 &- 3.62 \times 10^{-4}\, \Delta\beta - 2.58 \times 10^{-2}\, M_{w3} \\ &+ 6.39 \times 10^{-6}\, \Delta\beta^2 + 6.74 \times 10^{-2}\, M_{w3}^2 \\ &- 7.53 \times 10^{-8}\, \Delta\beta^3 - 4.30 \times 10^{-2}\, M_{w3}^3 \\ &- 2.38 \times 10^{-4}\, \Delta\beta\, M_{w3} + 1.45 \times 10^{-6}\, \Delta\beta^2 M_{w3} + 4.25 \times 10^{-5}\, \Delta\beta\, M_{w3}^2 \end{aligned} \tag{37}$$

eq. (37) is the loss of velocity along the blade passage through a theoretically impulse blade, relating the actual velocity to the theoretical velocity by $k_R = w_{exit}/w_{inlet}$.

$$P_v = \frac{1.85}{2}\,(1-\zeta)\,\rho_3 \left(\frac{n}{60}\right)^3 d_m^4\,(4.5\,H) \tag{38}$$

Finally, eq. (38) captures the losses due to partial admission. Weiss calls this loss ventillation losses, but it can also be called windage losses. Ventilation losses refers to the the non-admitted arc being dragged through stagnant air.

2.5 Turbine Blade Design

The design so far is based on 1D mean line theory. The design of the actual blade profile is what will guide the supersonic flow from the inlet direction to the outlet direction to maximize $c_{3u} - c_{4u}$. The basis of supersonic impulse blade profiling is often accomplished using an application of the Method of Characteristics (MOC) to transition a uniform flow field into vortex flow.

2.5.1 Method of Characteristics for Supersonic Impulse Blade Design

Vortex flow is a special type of supersonic flow in which every streamline follows a concentric path around a central point. This flow field was first discovered by A. Busemann [28], achieved by conditions where expansion and compression waves cancel each other to allow shockless rotation of the flow field. The vortex flow is bounded by the two circular arcs of the surfaces of which a desired surface Mach number is achieved. Boxer then first developed the method to transition uniform flow into vortex flow with MOC [9]. This method generates the profile for an inlet Mach number, turning angle and the mach numbers on the blade surfaces. Later analysis will need to be required to choose these desired Mach numbers.

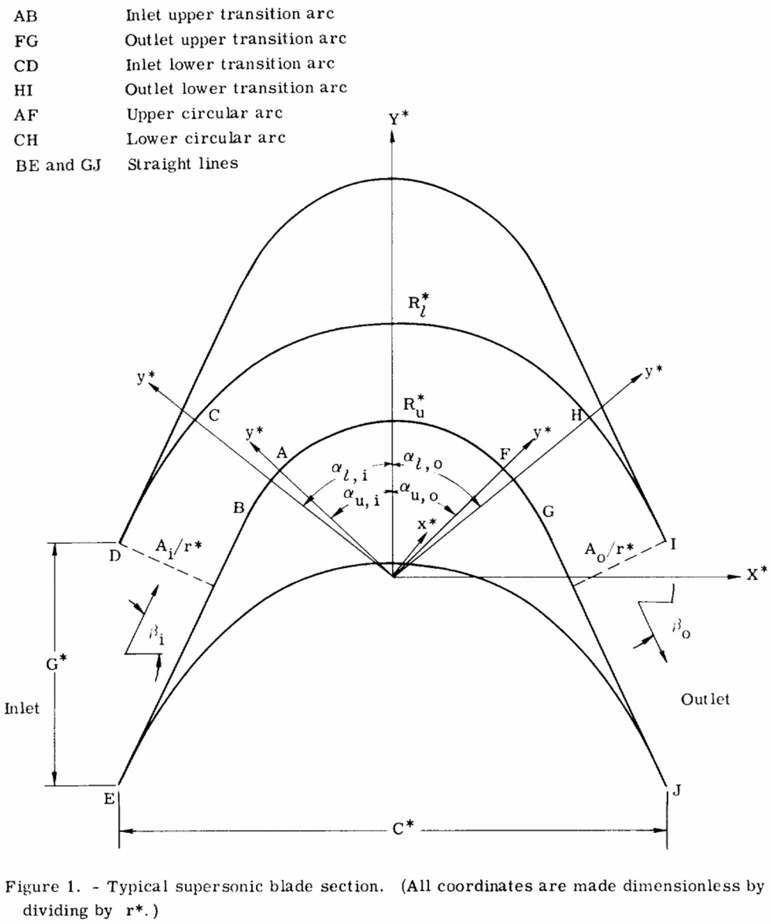


Figure 9: Typical MOC blade terminology.

Turbine blade terminology is based on the blade's upper and lower surfaces as shown in Figure 9. The top surface is the convex one, and the bottom surface as the concave one. This is in relation to the solid blades themselves as opposed to an alternate designation by the top and bottom of the blade passage. Supersonic vortex flow satisfies the equation:

$$cr = \text{constant} \tag{39}$$

Boxer [9] and Goldman [10] nondimensionalizes eq. (39) using the critical velocity $M^* = c/a^*$ where $a^* = \sqrt{\gamma RT^*}$ is the speed of sound at the sonic point. As a^* only depends on stagnation temperature, it is constant along any streamline of fixed stagnation enthalpy. Then, a nondimensional radius r^* is introduced as the sonic velocity streamline in the vortex field, giving the following condition:

$$M^* r^* = 1 \tag{40}$$

Goldman gives an equation [10] to find the Prandtl-Meyer angle from the critical mach number M^* as:

$$\nu(M) = \frac{\pi}{4}\left(\sqrt{\frac{\gamma+1}{\gamma-1}} - 1\right) + \frac{1}{2}\left(\sqrt{\frac{\gamma+1}{\gamma-1}} \arcsin\left[(\gamma-1)M^{*2} - \gamma\right] + \arcsin\left(\frac{\gamma+1}{M^{*2}} - \gamma\right)\right) \tag{41}$$

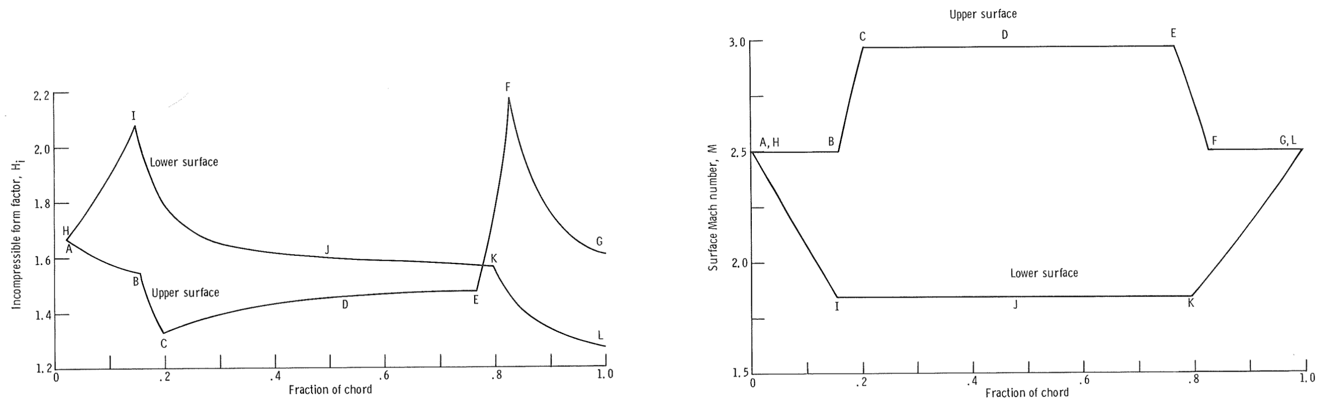
And the converting between Mach number and Critical Mach number is given by:

$$M^* = \sqrt{\frac{(\gamma+1)M^2/2}{1+(\gamma-1)M^2/2}} \tag{42}$$

The rest of the methodology to design the blade profile can be found in Goldman's paper [10] and will not be repeated here. Equations eq. (40) and eq. (42) are highlighted as they can be used to derive the nondimensional radius of the upper and lower surfaces of the blade, which will be important in the following discussion on surface Mach number selection. It is important to note that the inlet mach number for starting here is computed not with the absolute velocity c_3 but with the relative velocity w_3, as the flow will be turning inside the rotating frame of the turbine blade. This distinction is made as the relative velocity is dependent on the rotational speed of the turbine, meaning that this blade profiling is optimized for only the design point of the turbine. The surface Mach numbers are constrained by two mechanisms. Flow separation and supersonic starting.

2.5.2 Flow Separation

Separation occurs in high adverse pressure gradients. In supersonic flow, this happens whenever the flow is decelerating along the surface. There are two main regions on the blade surfaces which separation can occur. At the upper surface, the inlet transition arc accelerates the flow from the inlet velocity until the vortex flow, then decelerates the flow until the outlet. At the lower surface, the inlet transition arc decelerates the flow from the inlet velocity until the vortex flow, then accelerates the flow until the outlet. Goldman believes that separation on the lower surface is probably not as important as the flow would tend to reattach shortly as the pressure gradient becomes favourable again [1]. Goldman uses the value of the incompressible form factor H_i along the blade surface to predict separation, where Schlichting [29] found that separation usually occurs in the range of H_i from 1.8 to 2.4. The variation of H_i along the blade surface can be seen in fig. 10a.



(a) Incompressible form factor H_i along the blade surfaces **(b)** Distribution of Mach number along the blade surfaces

Figure 10: Goldman's published figures describing the distribution of Mach number and incompressible form factor H_i along the blade surfaces. [1]

Sasman and Cresci [30] provides a method to compute the development of the compressible turbulent boundary layer and thus find the incompressible form factor H_i along both surfaces. In this method, the edge Mach number distribution was taken from the MOC blade solution and fed into the their compressible integral method. The boundary layer was started as a turbulent flat plate with $\theta_0/c = 0.036\,\mathrm{Re}_{s_0}^{-1/5}$, using the prandtl $u/U = (y/\delta)^{1/7}$ and $\delta/c = 0.37\mathrm{Re}^{-1/5}$ [31]. The Reynolds number Re_{s_0} is calculated with the blade inlet density ρ_3, relative velocity w_3 and the chord length c. The shape factor $H_{i,0}$ is chosen to the value which the integration relaxes to within the first few percent of chord regardless of the start value. Once this calculation is done, the incompressible form factor H_i can then be used to evaluate if generated blade profile, and thus the chosen surface Mach numbers will lead to separation and where and feed into the design loop.
It is important to note that Goldman provides another set of equations in his NASA TN D-4421 1968 paper [10] that has been shown to not be conservative enough, with tests by Erikson on the development of the HM60 engine saying that his criteria is a little too optimistic [32]. The Sasman method described here is from Goldman's NASA TM X-2095 paper [1].

2.5.3 Supersonic Starting

On startup before the turbine passage runs supersonically, a detached bow shock stands ahead of the leading edges. The flow behind this normal shock is subsonic and has lost a significant amount of total pressure. For the turbine to operate nominally with supersonic flow, the passsage must be able to "swallow" this shock, allowing the shock to move through the whole passage.
To do so, the post shock flow with a reduced total pressure must have enough of a flow area pass the required mass flow similar to the choked flow condition. A similar condition is the unstart of a supersonic wind tunnel or a supersonic internal intake. However, Boxer notes that due to the large velocity gradient between the channel walls due to tue turning, the flow cannot be assumed one directional as are the case in the provided examples. Goldman provides

make s this ex plains stands

the condition [10] accounting for the "reduction of mass flow due to curvilinear flow" to calculate the maximum inlet design mach number to ensure starting as:

$$\frac{p_{02}}{p_{01}}(M_i) \;\geq\; \frac{Q}{1-C} \tag{43}$$

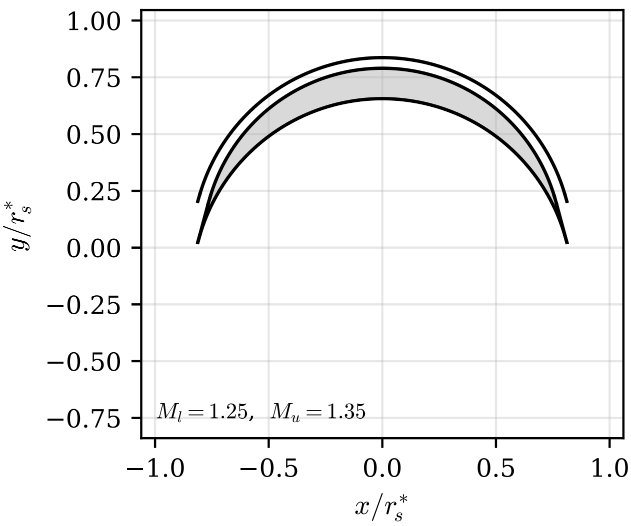
Where p_{02}/p_{01} is the total pressure loss across the normal shock, which is a function of M_i, the inlet design mach number. The right side of equation (43) involves nondimensional parameters Q and C which can be solved for through equation 27, 34a and 34b as a function of the surface mach numbers M_l^* and M_u^*. Q is named the vortex parameter [11] and C is the reduction in mass flow due to two-dimensional flow. Solving this equation for M_i gives the maximum inlet design mach number to ensure starting. Once again, this maximum inlet mach number is not based on absolute velocity c_3 but on the relative velocity w_3. The relative inlet mach number at the design point is lower than at startup, as $w_{3u} = c_{3u} - u$ following the velocity triangle shown in fig. 7. The starting condition at eq. (43) should be more conservatively evaluated at standstill to ensure that the turbine will start at all rotational speeds.
Historically, starting was a major problem for supersonic turbines. Before the theory was established, Colclough had to build a special jig in which every blade could be pivoted to vary the contraction until the blade started [28].

2.5.4 Surface Mach Number Selection

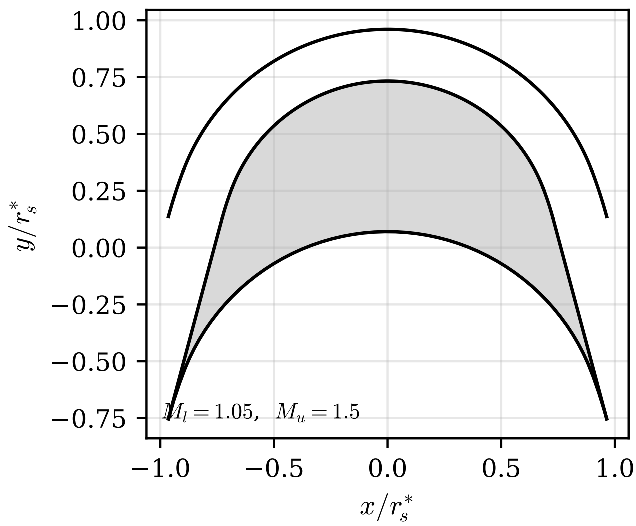
The separation criterion described in section 2.5.2 limits the acceleration of the flow along the upper surface, and the deceleration of the flow along the lower surface. Then, the starting criterion described in section 2.5.3 provides an additional limit on the inlet design mach number based on these surface mach numbers.
However, Goldman notes that for optimum efficiency, the maximum spread of surface number leads to maximum efficiencies [1]. It comes from the fact that a loss of velocity across the impulse blade leads to a loss of kinetic energy transfer to the blades, so he posits the efficiency of an impulse blade to be:

$$\eta = \left(\frac{w_{exit}}{w_{exit,ideal}}\right)^2 \tag{44}$$

eq. (44) is nearly the same as Weiss' eq. (37) loss. Weiss derived his empirical model only of 1D parameters such as inlet mach number and the angled the flow is turned. What this efficiency tries to capture is the boundary layer friction losses, shocks, and exit wake mixing losses. The effect of spreading the surface mach numbers becomes clear by looking at the vortex equation eq. (40). The larger the mach number spread, the larger the difference in R^* between the upper and lower surface, the larger physical passage width as shown infig. 11. The wider the channel compared to the blade chord, the lower fraction of the boundary layer thickness to passage width, the lower the wetted surface area per unit flow, and thus the lower the losses. This is the challenge of supersonic turbine blade design, maximum efficiency is achieved by riding along the edge of separation and starting.



(a) $M_l = 1.25$, $M_u = 1.35$



(b) $M_l = 1.05$, $M_u = 1.5$

Figure 11: The effect of surface Mach number spread on the blade passage, for the same inlet conditions. A narrow spread produces thin passage where the boundary layers occupy a large fraction of the width, while a wide spread opens the passage up.

2.5.5 Finite Leading Edge

In section 2.5.1, the upper and lower surface are mathematically designed to produce no shocks in transitioning the unifrom flow from the nozzle exit into the vortex flow. For no shocks to form, both surfaces must be perfectly parallel

to the starting flow. This is problematic because this implies the point where the two surfaces intersect at the leading edge, the angle between the upper and lower surfaces is zero. This creates an impossible to manufacture zero thickness leading edge. Boxer suggests a few methods to solve this issue. He separates design into turbines with a subsonic or supersonic axial velocity component.
Boxer [9] describes a few finite leading edge treatments depending on the axial velocity component. For subsonic axial velocity turbines, either lower or upper surface can be designed with an incidence angle to the inlet flow. Angling the upper convex surface and keeping the lower concave surface parallel to the inlet flow, the compressive oblique shock will be generated along the region between the inlet and the nozzle exit, affecting the flow in the next blade passage. Kantrowitz shows that in steady state a compression shock of equal strength will be created to obtain the steady state condition [33]. To solve this, the lowerer concave surface can be angled so that the shock stays within the blade passasge and does not affect the upstream flow as shown in fig. 12. The oblique comrpession shock can then be designed to intersect the upper surface at an angle to fully cancel out the shock. For supersonic axial velocity turbines, a wedge can be designed on the convex surface with a low enough angle from the design inlet flow so that the shock stays within the blade passage and can be cancelled out by a kink on the concave surface.

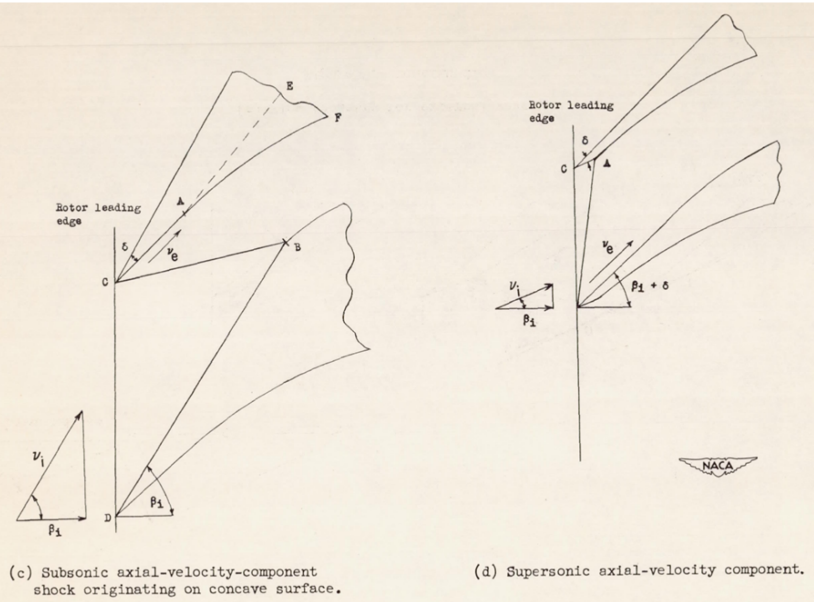


Figure 12: Finite leading edge treatments for supersonic blades: incidence angling of one surface, in passage shock cancellation, and the small wedge used to cancel the resulting compression shock.

2.5.6 Nozzle Design

The nozzle is sized with the isentropic choked mass flow equation eq. (45).

$$\dot{m} = \frac{A_{th}\,p_{01}}{\sqrt{T_{01}}}\sqrt{\frac{\gamma}{R}}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{45}$$

While the throat sets the mass flow, the ratio of the exit area to the throat area sets the exit Mach number, through the isentropic area relation eq. (46).

$$\frac{A_3}{A_{th}} = \frac{1}{M_3}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_3^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{46}$$

Historically, the optimal nozzle design for a ideal inviscid gas was designed with the Method of Characteristics as well. Rao provides a method to achieve this [34]. NASA special pubplication 8120 explains that a single canted parabola closely approximates the ideal nozzle contour and should be used for high performance nozzles [2]. In this thesis, the nozzle was designed as a simple conical nozzle with a 7° degree half angle. It can be seen from fig. 13 that the performance of a conical nozzle for low area ratios lead to a efficiency loss below 2%.

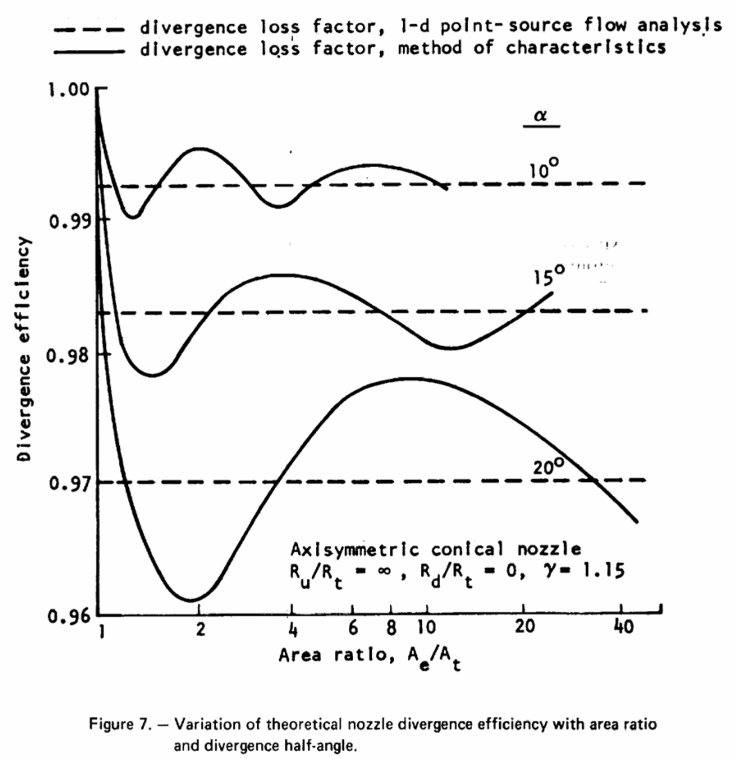


Figure 13: Calculated efficiency of a conical nozzle as a function of area ratio from NASA SP-8120 [2]

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3 Design

The turbopump design point for this thesis was chosen from the capabilities of the test rig. The design parameters for the pump and turbine is shown in table 1 and this section explains how each number was chosen. The RPM limit was set by the maximum speed of the Magtrol TM306 inline torquemeter and the AHB-3 hysteris brake at 20,000 RPM. The pressure rise and flow rate was chosen to study the low specific regime. The supply pump was decided to be Hozelock's JET 3000 K7 Jet pump as it was available in the department and had a published performance curve. At 0.3 kg/s, the supply pump had a head of around 2.2 m. Then, the pressure rise of the pump was chosen to be 20 bar for a specific speed of 6.4.

Table 1: Turbopump system design point.

Symbol	Parameter	Value
n	design shaft speed	20 000 rpm
Q	pump design volume flow rate	0.3 L/s
H	pump design head rise	204 m
Δp	pump design pressure rise	20 bar
n_q	pump specific speed (rpm, m^3/s, m)	6.4
P	turbine design shaft power	2.6 kW
p_{01}	turbine design inlet total pressure	7.65 bar
–	pump working fluid	water
–	turbine working fluid	air

3.1 Pump Design

3.1.1 Sizing

Following the global design point parameteres, the pump was sized using the Barske as described in section 2.3.2 to give the dimensions shown in table 2 and labelled by fig. 14.

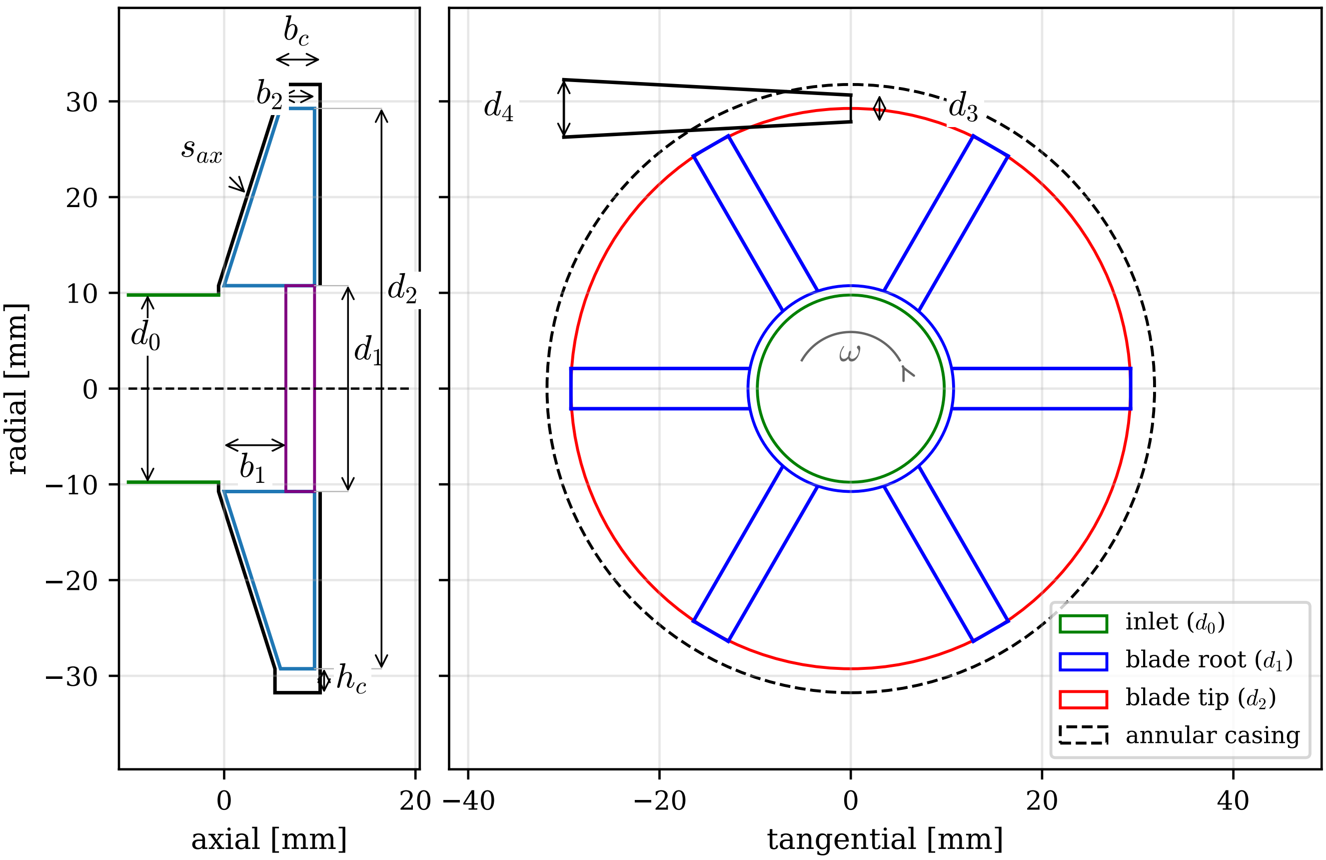

Figure 14: The sized pump geometry, drawn to scale from the sizing code output, with the meridional section (left) and end view (right) sharing the radial axis. The meridional section shows the open tapered blades, the conical front casing maintaining the axial clearance s_{ax}, and the annular casing of width b_c and depth h_c. The end view shows the six radial blades and the single tangential conical diffuser with throat d_3 and exit d_4. Values are listed in table 2.

The design starts following section 2.3.2 from what Barske considers a reasonable inlet velocity. 1.0 m/s was chosen. Higher inlet velocities lead to higher NPSHR. This fixes the eye diameter d_0. The eye is then enlarged to the blade leading edge diameter d_1 to allow the deliberate prerotation Barske recommends. The inlet and exit blade heights follow from continuity following eq. (16). The impeller tip diameter d_2 comes from the recovery equation eq. (15) with a deliberately conservative diffuser recovery coefficient of $k_d = 0.2$. A conservative k_d oversizes the impeller to make sure the desired head is achieved and can be adjusted in future iterations. The sized values are listed in table 2.
This geometry is then put into Lock's model to predict the head curve fig. 32 and NPSHR fig. 36 to be compared with experimental data in section 5.

Table 2: Pump design summary. The impeller dimensions are the sizing code output; the diffuser throat d_3 and exit d_4 are the as-built values, which depart from the sized values for the reasons given in the text.

Symbol	Parameter	Value
d_1	impeller inlet (blade leading-edge) diameter	21.5 mm
d_2	impeller tip diameter	58.5 mm
b_1	blade height at impeller inlet	6.4 mm
b_2	blade height at impeller exit	3.6 mm
z	blade count	6
d_3	diffuser throat diameter (as built)	3.0 mm
d_4	diffuser exit diameter (as built)	9.0 mm
ψ	design head coefficient	1.065
n_q	specific speed (rpm, m^3/s, m)	6.4

3.1.2 Diffuser

The diffuser is a single conical passage placed tangentially on the annular casing, following Barske's guidance from section 2.3.2. The throat was sized for a throat velocity of 0.8 to 0.9 of u_2, giving $d_3 = 2.79$ mm, and the exit area was set at 3 to 4 times the throat area, giving $d_4 = 5.58$ mm. Two changes were made to these before the part was made. The throat was opened up to 3.0 mm, rounding the sized value up. The oversize itself is for manufacturing,

so the small passage prints and cleans out reliably, but rounding up rather than down was deliberate. The throat is the most cavitation critical dimension in the pump, and a larger throat runs a lower throat velocity, which raises the cavitation limited maximum flow that the results in section 5 show this throat actually sets. The low specific speed optimisation literature points the same way, treating the diffuser throat as a lever to open rather than to minimise [7]. The diffuser exit was then enlarged to 9.0 mm for packaging. The cone sits tangentially on the annular casing, and had to clear the casing wall for the inlet fitting. That over-expands the passage well past Barske's 3 to 4 area ratio guidance, but the extra area is all downstream of the throat on the discharge side, so it costs nothing here. The comparison of this throat sizing against the literature is discussed in section 5. what?
Barske comments that the number of impeller blades does not seem to be critical, with 3-6 being operated successfully. However, altering the number of blades can be used to elminite fluid vibrations [17]. Lock provides corrections for blades from 1 to 16 [15].

3.1.3 Mechanical Design

After the theoretical Barske pump geometry has been sized to deliver the design head at a desired flow rate, it has to then be translated into a manufacturable assembly that satisfies additional practical requirements. The mechanical design has to survive loading conditions such as high rotational speeds and pressures. Then, it also has to be leak tight, manufacturable and assemblable. A CAD model of the pump assembly that satisfies these requirements is shown in fig. 15.

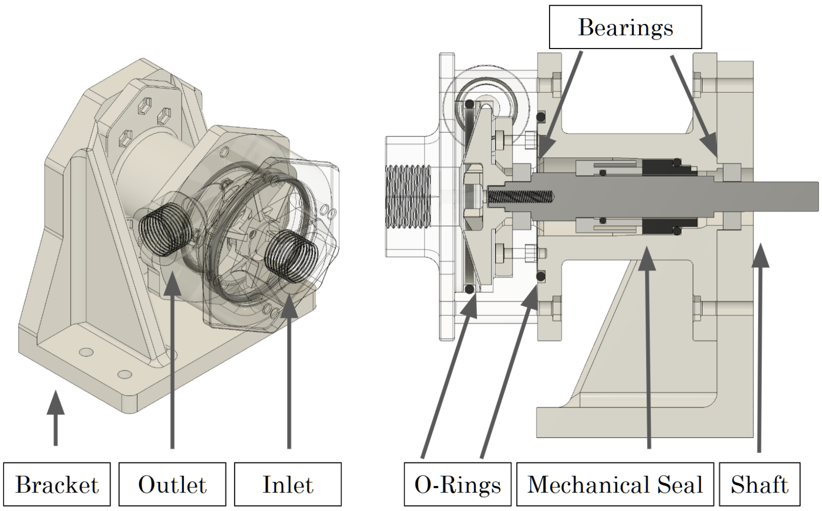

Figure 15: Labeled CAD isotropic and section view of the pump assembly

Material Selection The low pressure rise of 20 bar as a small scale prototype to study rocket engine pumps allowed for a wider and more accomodating material selection criteria. Plastic was chosen for its ability for rapid prototyping and low cost. SLA was chosen instead of FDM for most parts because FDM printing layer by layer creates a pourous surface that will not be leak proof at 20 bar. The pump impeller and were printed in Formlabs Tough 2000 resin for this reason. For this thesis, it was realized that although mechanically weaker than Tough 2000, the Formlabs clear resin was found to be a possible material. The pump casing and inlet were printed in clear resin and polished. This allowed for cavitation visualisation in section 5, which turned out to be one of the most valuable measurements of the project. For the bracket that mounted the pump to the test rig, leakage was not a concern so it was printed in PLA for its rigidty and low cost. The pump shaft had to be made of a stronger material for rotordynamic stability at high rotational speeds and high torque, so it was made of stainless steel and machined on the manual lathe and mill in the Imperial Advanced Hackspace.

Sealing Most rotary seals did not work for this application. A lip seal wears out fast at this surface speed and holds very little pressure, a labyrinth seal could not be reliably used with plastic parts due to printing tolerances, and packing has too high friction. This led to the choice of using a mechanical seal. Mechanical seals consist of a spring loaded rotating face running against a stationary face with the only leak path a thin liquid film between two lapped surfaces. The liquid film greatly reduces the friction between the two faces. A John Crane 59U-14 mechanical seal was used. Mechanical seals operate by pressing a rotating face against a stationary face with a thin liquid film in between [35]. For the static seals, two O-rings sized with the [36] were used to seal the inlet against the casing and the casing to the bearing housing. The seal between the inlet and the casing was a piston seal, while the seal between the casing and the bearing housing was a face seal.

Mechanical Losses The mechanical losses of this pump assembly come from the bearings and the mechanical seal. The bearings were 61900 series stainless steel ball bearings chosen for their water resistance and cost. In this design, one bearing is submerged in water, increasing the drag loss. The power loss of the bearing was estimated by SKF's frictional moment model [37]. Karrasik gives a formula for mechanical seal friction [35] with eq. (47).

$$\tau_{seal} = p_f \, A_f \, r_m \, f, \qquad P_{seal} = \tau_{seal} \, \omega \tag{47}$$

where B_r is the seal balance ratio built from the mechanical seal face outer and inner diameters d_o, d_i and the balance diameter d_b as shown in fig. 16, $k \approx 0.5$ is the pressure profile factor for the fluid film between the faces assuming a linear pressure distribution, and p_{sp} is the spring force over the face area and Δp is the pressure differential across the seal face [4].

$$p_f = \Delta p \, (B_r - k) + p_{sp}, \qquad B_r = \frac{d_o^2 - d_b^2}{d_o^2 - d_i^2} \tag{48}$$

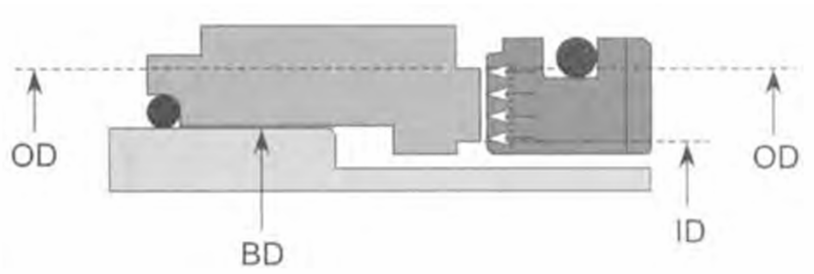

Figure 16: Geometric dimensions of an OD pressurized the mechanical seal by Karrasik [3].

In conventional shrouded impellers, the back face of the impeller is exposed to the the full pressure rise of the pump, contributing to a higher pressure Δp that needs to be sealed across the shaft, leading to higher mechanical seal friction. The Barske impeller is open and the back face is exposed to the suction pressure, which reduces the pressure differential across the seal and therefore the friction.

Concentricity Concentricity is of great importance in high speed rotating machinery. The design takes advantages of shoulders and press fits to ensure the concentricity between every interfacing part as shown in fig. 15.

3.2 Turbine Design

3.2.1 Design Point and Regime Shift

Turbine sizing has to satisfy many requirements simultaneously to reach the global requirements of a rotational speed of 20 000 rpm at a power of 2.6 kW. Parameters such as diameter d_m, mass flow through the turbine $\dot{m}$, blade inlet angle β_3, pressure ratio p_{01}/p_{02} have to be balanced against each other to reach a feasible design point.
The turbine has two operating regimes, and the distinction runs through this section and the results. The design point is 2.6 kW at 20 000 rpm on 7.65 bar supply. The test point is what the laboratory shop air could actually deliver, around 2.9 bar absolute at the stator with the pressure drooping under flow, which settled the coupled shaft at around 5400 RPM. table 3 compares the two. The choked supersonic nozzle behaves the same in both regimes because choked flow is speed independent, but the rotor stage similitude parameters were not matched at test, and this difference drives most of the discussion in section 5.

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Table 3: Turbine design point against the achieved test point. The test supply pressure drooped under flow, so the quoted value is representative.

Parameter	Design	Test
Shaft speed	20 000 rpm	≈5400 RPM
Inlet total pressure	7.65 bar	≈2.9 bar
Inlet total temperature	300 K	≈308 K
Rotor inlet relative Mach M_{w3}	1.33	1.58
Blade to jet speed ratio	0.214	≈0.06
Rotor state	started	unstarted

3.2.2 Architecture Selection

The established rocket turbopump turbine is an axial flow stage, to the point that the standard NASA design monograph for these turbines treats only axial machines [24], so the choice here of a radial inflow cantilever stage needs justifying. In a radial inflow cantilever, also called a centripetal impulse turbine, the flow enters the rotor radially inward through blades cantilevered off the rim of the disc, instead of passing axially through a bladed wheel. The reason for it is manufacturing and integration. At this scale the whole rotor is a single printed part overhung on the common pump shaft, with no separate turbine bearing, gearing or coupling, and a small nozzle block feeds a short arc of the rim through a compact manifold. An axial wheel would need axial length and a hub that fights the overhung single shaft layout, and is harder to make at good efficiency at this size [38]. The radial inflow cantilever is the layout proven for small supersonic turboexpanders in this exact power and manufacturing regime [20], and for a compact supersonic rocket turbine of the same kind [39]. It costs a few points of efficiency against an axial or a radial inflow reaction stage at the design point [39], which is an acceptable trade for a single stage that is already far off its optimum speed ratio. The supersonic impulse blade itself is the same object either way, it is an axial impulse profile wrapped onto a radial rim [39], and the meanline loss model is the same for the axial and the radial inflow cantilever stage [20], so this is a packaging choice and not a change to the aerodynamic design method.

The stage is an impulse stage, for the reasons established in section 2, and its tolerance to tip clearance matters here because the rotor is printed [24]. The central constraint is diameter. Running the stage at the optimum blade to jet speed ratio on a 20,000 RPM shaft would need a mean diameter in the hundreds of millimetres, which does not fit the rig or the shaft dynamics, so the turbine accepts a small diameter and a heavily off optimum speed ratio instead. The small mass flow then cannot fill the annulus at any sensible blade height, so partial admission is used. The degree of admission $\varepsilon = 0.15$ was chosen by working eq. (34) backwards, picking the admission fraction that raises the blade height to 4.0 mm, the smallest height that prints reliably and keeps the aspect ratio sensible.

3.2.3 Sizing

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The mean line sizing is the inverse design chain of the theory chapter, executed in order. The inputs are the shaft power P, speed n, mean diameter d_m, mass flow $\dot{m}$, nozzle angle β_3, degree of admission ε, the gas properties and the exit pressure p_e. The blade speed follows from the speed and diameter, $u = \omega d_m/2$, and the required specific work from the power, $\Delta h_{\text{use}} = P/\dot{m}$. The velocity triangle inversion eq. (30) then fixes the jet the nozzle must deliver, c_{3u}, c_3 and c_{3m}. The nozzle exit static state follows from conservation of energy in the adiabatic nozzle and the isentropic relations:

$$T_3 = T_{01} - \frac{c_3^2}{2c_p}, \qquad a_3 = \sqrt{\gamma R T_3}, \qquad M_3 = \frac{c_3}{a_3} \tag{49}$$

$$p_3 = \frac{p_{01}}{\left(1 + \frac{\gamma-1}{2} M_3^2\right)^{\frac{\gamma}{\gamma-1}}}, \qquad \rho_3 = \frac{p_3}{RT_3} \tag{50}$$

Because the stage is impulse, the rotor takes no static pressure drop, so the nozzle exit static pressure must equal the exit pressure p_e. This is a modelling simplification worth stating. A radial inflow cantilever is not strictly an impulse stage, because the flow moves radially inward through the rotor inside the blade centrifugal field, which puts a small static pressure drop across the rotor and a corresponding degree of reaction, and this is why the class is properly called quasi impulse [20]. Here the blade speed is far below the jet velocity, the same off optimum condition that sets the efficiency, so the centrifugal reaction is small, and the stage is sized as pure impulse throughout with the reaction neglected in both the velocity triangles and the loss model. The near optimum cantilever of Weiss carried a reaction of about 0.13 [20], so the simplification is justified only by the low blade to jet speed ratio of this design, and it is carried as a known limitation of the design model. The supply pressure p_{01} is therefore not free: it is iterated until $p_3 = p_e$, which is how the design tool arrives at the required stator supply pressure of 7.65 bar rather than assuming it. The blade height then follows from continuity through the admitted arc, eq. (34), and the total nozzle throat area from the choked flow relation eq. (45). Finally the Weiss loss chain of section 2.4.5 is wrapped around this ideal chain in an outer loop: the internal power is raised until the shaft power net of the rotor loss and ventilation meets the 2.6 kW target, so the quoted supply pressure already carries the losses.

With the chain in place, the remaining free choices are the mean diameter and mass flow. These were chosen by sweeping the $(d_m, \dot{m})$ plane with the full model and overlaying the manufacturing and rig constraints: a minimum blade height of 4 mm for reliable printing, a minimum nozzle throat width of 0.5 mm, and a nozzle exit Mach ceiling of 1.8 to keep the rotor inside the validated range of the loss correlations. fig. 17 shows the swept design space, the feasible region, and the selected design point. The chosen $d_m = 95$ mm and $\dot{m} = 0.0428$ kg/s sit inside the feasible region with the blade height constraint binding, which is the quantitative version of the statement that the blade height, not the aerodynamics, sizes this machine.

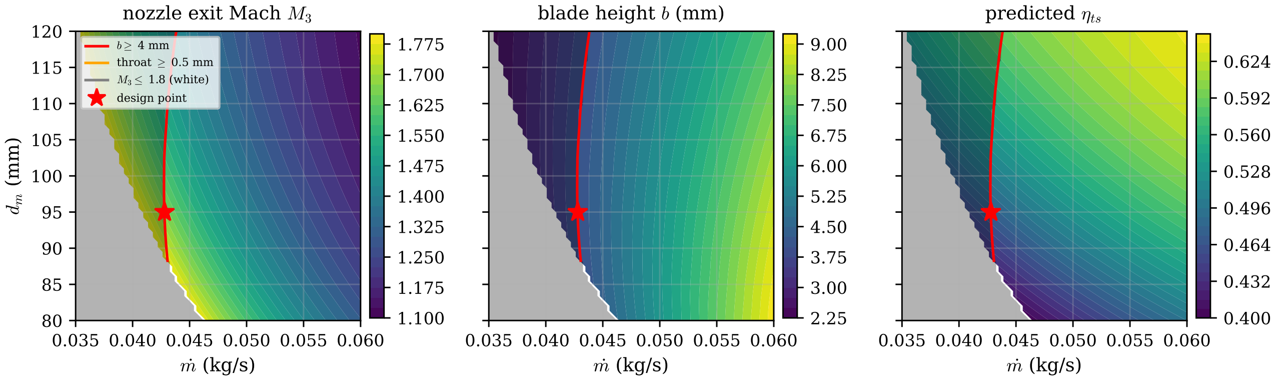


Figure 17: The turbine design space over mean diameter and mass flow, evaluated with the full sizing and loss model at the design speed. The panels show nozzle exit Mach number, blade height and predicted total to static efficiency. Overlaid lines are the design constraints (blade height $\geq$ 4 mm, nozzle throat $\geq$ 0.5 mm, $M_3 \leq 1.8$), the shaded region is infeasible, and the marker is the selected design point.

The mean line sizing gives a mean diameter of 95 mm, a blade speed of 99.5 m/s at design speed, a nozzle exit velocity of 464 m/s at Mach 1.67, and a rotor inlet relative Mach number of 1.33. The full output is in table 4 and the design point velocity triangles are drawn to scale in fig. 18. The triangles make the off optimum operation visible directly, as the blade speed is small next to the jet velocity and the exit swirl is strongly negative, which is the kinetic energy the stage fails to extract at this speed ratio.

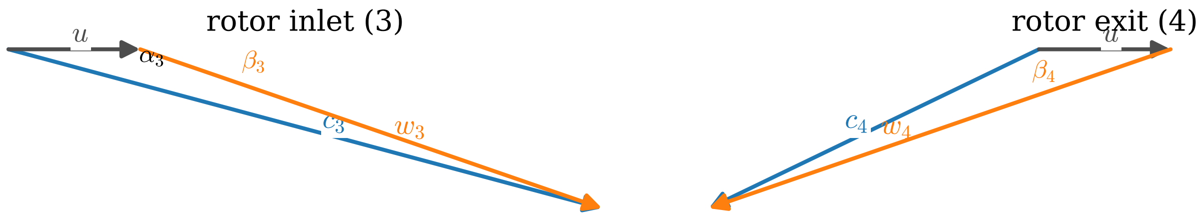


Figure 18: Design point velocity triangles at the rotor inlet and exit, drawn to scale from the sizing output. The exit triangle is the ideal symmetric impulse case. The large negative exit swirl is the direct consequence of operating at a blade to jet speed ratio of 0.214, far below the optimum.

Table 4: Turbine design summary, generated directly from the design code.

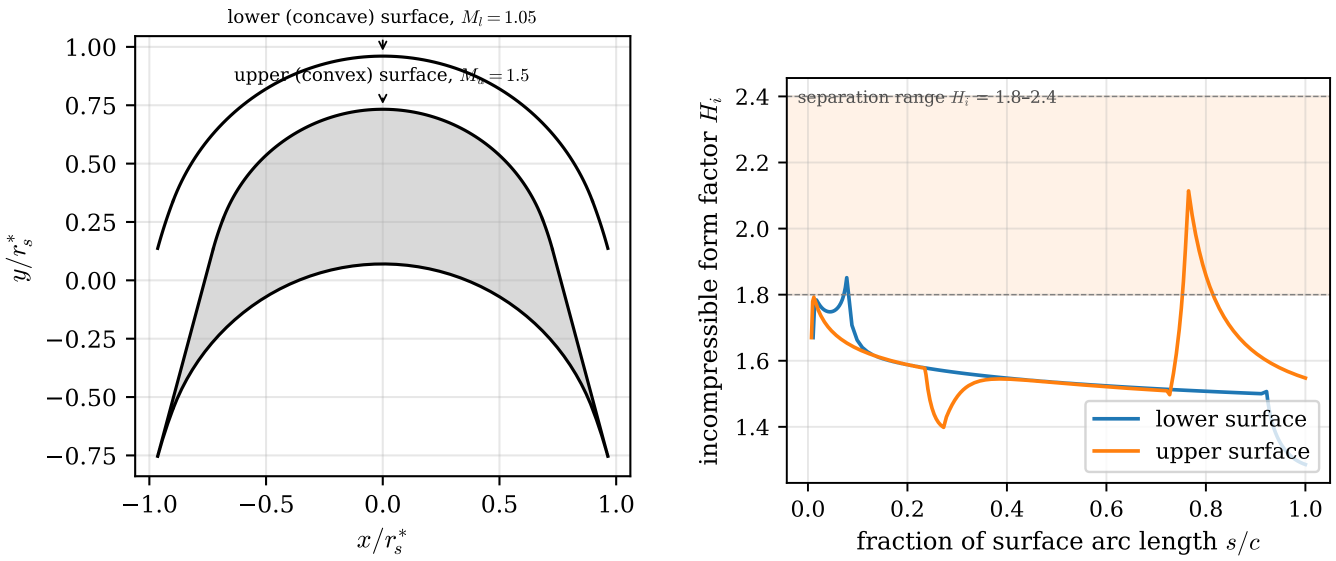
Symbol	Parameter	Value
P	design shaft power	2.6 kW
p_{01}	design inlet total pressure	7.65 bar
T_{01}	design inlet total temperature	300 K
$\dot{m}$	design air mass flow rate	0.0428 kg/s
d_m	mean (pitch) diameter	95 mm
b	rotor blade height	4.0 mm
ε	degree of admission	0.15
β_3	rotor blade angle (from tangential)	15°
z_N	number of nozzles	3
A_{th}	total nozzle throat area	35.5 mm^2
c_3	design nozzle-exit velocity	464 m/s
M_3	design nozzle-exit Mach number	1.67
M_{w3}	design rotor-inlet relative Mach number	1.33
η_{ts}	design total-to-static efficiency	0.47

3.2.4 Nozzle

The ideal nozzle contour for this application would be a Rao or canted parabola design as described in section 2. Due to time constraints and the negligible performance difference at this low area ratio, a simple conical nozzle with a 7 degree half angle and rounded corners was used instead. NASA SP-8120 puts the efficiency penalty of a conical contour below 2% at low area ratios, which is small against the other losses in this stage, so this is a sensible engineering trade and not a compromise of the results. There are 3 nozzles with a total throat area of 35.5 mm^2.

3.2.5 Blade Passage Design

The blade passage was generated with the MoC procedure of section 2.5.1 for an inlet relative Mach number of 1.3 at a blade angle of 75 degrees from axial, with surface Mach numbers $M_l = 1.05$ and $M_u = 1.5$. The choice of this surface Mach pair is the central design decision of the blade. The pair is not free: their mean is fixed near the inlet Mach number and their spread sets the turning the stage needs. The lower bound is separation on the pressure surface, checked with the Sasman and Cresci method of section 2, and the upper bound is starting, checked with the Goldman criterion derated by the Colclough fit. Following Goldman's observation that efficiency rises with Mach spread, the spread was opened until separation became imminent. fig. 19 shows the result: the as-designed blade with the H_i check touching 2.1 at around 77% chord on the upper surface, deliberately separation imminent at the design point. The boundary layer displacement correction applied to the printed contour is shown in fig. 20.
The starting margin was left deliberately thin to maximise loading. The Goldman limit for this passage is $M_{w3} = 1.408$ against a design value of 1.33. The same calculation run off design predicted, before any coupled test took place, that the rotor cannot start below approximately 15 300 RPM, because at low blade speed the relative inlet Mach number rises above the limit. This prediction is a design stage statement. What happened when the coupled test ran far below this threshold is the subject of section 5.



(a) As-designed blade passage

(b) Separation check

Figure 19: The as-designed MoC blade passage and its separation check. The form factor H_i is evaluated on both surfaces at the design point inlet conditions and touches 2.1 at around 77% chord on the upper surface, at the bottom of the 1.8 to 2.4 separation band. The blade is separation imminent by design, which per Goldman is where the efficiency optimum sits.

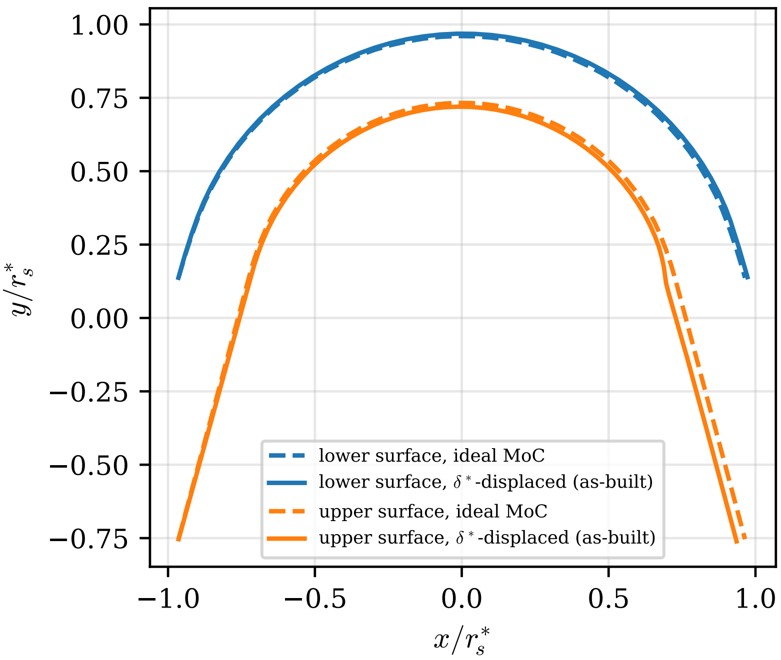


Figure 20: Ideal MoC contour (dashed) against the boundary layer displaced contour that was actually manufactured (solid). The surfaces are offset by the local displacement thickness δ^* from the Sasman and Cresci solution, so that the printed passage presents the inviscid design geometry to the core flow.

3.2.6 Losses and Predicted Efficiency

Applying the Weiss loss chain of section 2.4.5 at the design point gives a nozzle velocity coefficient of $k_N = 0.917$, a rotor velocity coefficient of $k_R = 0.807$, and a ventilation power of 85 W. Together these take the ideal stage efficiency of around 65% down to a predicted total to static efficiency of 0.47. Whether that prediction holds is taken up against the measured loss data and the literature in section 5.

3.2.7 Blade Geometry Details

Three geometry details depart from the ideal MoC contour. First, the leading edge. The MoC construction produces a zero thickness leading edge, which cannot be printed. A finite wedge of about 5 degrees was added on the pressure surface following the approach in section 2. The shock detachment angle at the design relative Mach number of about 1.3 is around 6.5 degrees, so a 5 degree wedge keeps the oblique shock attached. Second, the trailing edge is truncated to a flat of about 0.8 mm so it survives both the shock loading and embrittlement, since the expansion cools the air to around −88 °C at the exit. Third, balance holes are drilled through the disc web to equalise the pressure across the rotor and remove around 400 N of parasitic axial thrust. Balance holes and pistons are standard rocket pump practice [40], and a CFD study of a Barske type pump in the same regime showed balance holes reducing the axial force from 17 kN to 350 N, the same order of effect as needed here [5]. Finally, the blades are shrouded, which reduces the blade tip deflection by a factor of around 48 and removes tip leakage as a loss mechanism.

3.2.8 Cold Flow Thermal Environment

Expanding room temperature air to Mach 1.67 cools it to a static temperature of around 185 K, and the printed resin blades have to survive this. The static temperature is however not what the blade surface sees. A surface in high speed flow recovers part of the kinetic energy as frictional heating in the boundary layer, and equilibrates at the adiabatic wall temperature

$$T_{aw} = T_3 + r\,\frac{w^2}{2c_p}, \qquad r = \mathrm{Pr}^{1/3} \tag{51}$$

where r is the recovery factor for a turbulent boundary layer and w is the local relative velocity [?]. With $\mathrm{Pr} \approx 0.70$ for cold air, $r \approx 0.89$, and at the design rotor inlet relative velocity the recovery term adds back around 58 K, so the blade surfaces equilibrate near −30 °C rather than the static −88 °C.

Two follow up checks close the question. First, the blade thermal time constant. Treating the thin blade as a lumped capacitance with the film coefficient from a turbulent flat plate correlation evaluated at the blade chord, the time constant $\tau_{th} = \rho_b\,(t/2)\,c_{p,b}/h$ comes out below one second, so the blades reach their steady temperature within seconds of the start of a run and there is no transient grace period to exploit. Second, dimensional stability: the thermal strain at 50 K below ambient with the resin's expansion coefficient of order 10^{-4} per K is about 0.6%, which over the 4 mm blade height is some 25 µm, small against the running clearances. The thermal design conclusion is therefore that the geometry is unaffected but the material toughness at −30 °C is the binding concern, which is what motivated the truncated trailing edge flat of the previous section.

3.2.9 Structural and Rotordynamic Analysis

The rotor was checked with static FEA at the design speed: blade root bending under the combined centrifugal and gas loads, blade deflection, shroud hoop stress and disc bore stress. The blade bending safety factor is 39 on the Tough 2000 material. All FEA was run in Fusion 360 static studies. For linear elastic statics the solver brand is irrelevant, and each result was cross checked against a hand calculation of the matching idealised case, which is the verification that actually matters.
The hand calculations are the classical rotating component results. The peak hoop stress of a solid rotating disc bounds the disc and bore stress,

$$\sigma_{\theta,max} = \frac{3+\nu}{8}\,\rho_b\,\omega^2 r^2 \tag{52}$$

the centrifugal pull at the blade root is the integrated body load of the blade above it,

$$\sigma_{root} = \frac{\rho_b\,\omega^2}{2}\left(r_{tip}^2 - r_{root}^2\right) \tag{53}$$

and the shroud ring carries its own centrifugal load as a free hoop,

$$\sigma_{hoop} = \rho_b\,u_{shroud}^2 \tag{54}$$

where ρ_b is the blade material density and u_{shroud} the shroud rim speed. The gas bending load on each blade adds a cantilever bending stress $\sigma_b = M_b\,y/I$ at the root, with M_b the bending moment from the blade force of the velocity triangles and y the extreme fibre distance. Each FEA peak stress was required to agree with its idealised counterpart to the level the idealisation allows, which catches unit, constraint and load errors that mesh refinement cannot.
For the shaft, the finite element rotordynamics of fig. 22 is cross checked by the overhung rotor estimate. Treating the shaft as a cantilever of bending stiffness EI with the disk mass m at radius L from the bearing,

$$\omega_1 \approx \sqrt{\frac{3EI}{mL^3}} \tag{55}$$

which lands in the same regime as the finite element first bending pair and confirms the order of the margin.
For rotordynamics, the shaft is a 10 mm solid aluminium stub. A finite element rotor model with the turbine disk overhung and the measured disk mass and inertias places the first bending natural frequency at standstill around 99,000 RPM, and the Campbell diagram in fig. 22 shows the margin over the full operating range. The first synchronous crossing sits far above the 20,000 RPM design speed, with a margin of better than 3.5. The shaft connects to the pump through a jaw coupler with Double-D 7 mm flats, sized with a compressive safety factor of around 10 on the printed hub.

Figure 21: Rotor static FEA at 20 000 rpm: blade root bending, shroud hoop stress and disc bore stress, with the resulting safety factors.

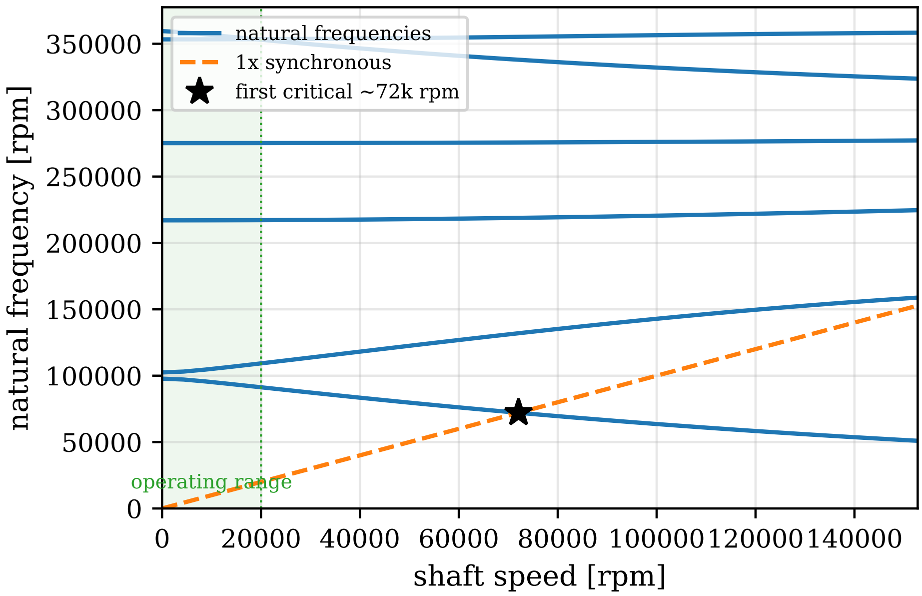


Figure 22: Campbell diagram of the turbine shaft from a finite element rotor model (10 mm solid aluminium shaft, turbine disk overhung, ball bearing stiffness; the coupler side disk is omitted). The first bending pair starts near 99,000 RPM at standstill and the backward whirl branch crosses the synchronous line at around 72,000 RPM, leaving a margin of more than 3.5 over the operating range.

3.2.10 Manufacturing

The rotor and the stator nozzle block were printed by SLA in Tough 2000 resin, washed in IPA, UV cured, and the mating faces sanded flat. Printed resin turbomachinery at this power level is published practice rather than an improvisation, as Weiss cites both an experimental kW scale SLA resin air turbine and SLA diffuser vanes in operation [20].

4 Experimental Setup

The experimental setup was designed for three specific configurations: Electric motor driven pump testing, Isolated turbine coupled to a hysteris brake, and integrated turbine and pump testing.

4.1 Test Configurations

(a) Motor driven pump **(b)** Turbine against brake **(c)** Coupled turbopump

Figure 23: The three test configurations. In each case the TM306 torquemeter sits between the driving component and the load, so torque is measured at the shaft.

In all three configurations, there is a driving component, the torquemter and a load. fig. 23a shows the electric motor driven pump testing configuration. The electric motor test is used to gather most of the data because the motor is more controllable, repeatable and safer. In this configuration. The second configuration in fig. 23b connects the turbine to a hysteris brake to test the turbine in isolation. Due to time constraints, this configuration was not tested. The final configuration in fig. 23c connects the turbine and pump together to test the performance of the integrated system.

4.2 Mechanical Layout and Shaft Train

At the design 20,000 RPM for the rig, shaft alignment, rigidity and maintainability were the main drivers. At 20,000 RPM, available shaft couplers had barely any tolerance for misalignment. The Magtrol MIC-5-2470 couplers had a maximum 0.8mm axial, 0.36mm radial and0.7° angular misalignment tolerance at 35,000 RPM. Shaft couplers had to be used to connect rotary components to the transducers, a requirement that doesn't have to be satisfied by turbopump units as the shaft can be machined as a single piece.
The central component of each configuration in fig. 23 is the Magtrol's TM306 inline torquemeter as it allows the stated ¡0.1% torque measurement accuracy. As the torquemeter was designed to be mounted onto a base plate, a base plate design was chosen instead of an integrated assembly like a normal turbopump.
The base plate is a 20 mm plywood with a laminate on both sides to be semi-waterproof, to have a flat surface for mounting, rigid but also cheap. M6 and M8 Countersunk holes were drilled into the base plate with a CNC router to allow for precise mounting of the components.
Each rotating component sits on its own mount bolted down to this baseplate. The mounts were 3D printed with a flat rectangular base and the same 80.00 mm shaft height as the torquemeter, so that bolting a mount flat to the baseplate already puts its shaft on the torquemeter centreline in the vertical direction without any shimming. The mounts also carry the alignment reference faces described below. The whole turbomachinery stand is then surrounded by a blast shield, a frame of aluminium extrusion carrying a 20 mm Dyneema panel on the faces looking at the rotating parts and a 15 mm polycarbonate window where the test has to be watched. The frame is bolted to the bench rather than to the baseplate, so that a failure of the rig does not feed straight into the shield through the same structure. The sizing of these panels is the subject of the protection system design later in this chapter.

Figure 24: The assembled test rig with the major components labelled: motor, couplers, torquemeter, pump, turbine and the throttle valves on the CNC drilled baseplate.

4.2.1 Shaft alignment

The reason the shafts had to be aligned was because flexible shaft couplers were used. These tolerate some misalignment at high speeds, an essential requirement for a test rig where two or more shaft have to be mechanically connected.
The alignment of each component to the torquemeter had many challenges. Mounting magnetic dial indicators onto the shafts was not possible due to three reasons. The shafts were made of austenitic stainless steels so they were non-magnetic, the shafts were short and had a low diameter (10-15mm diameter, with maximum stub length of 20mm). A short stub length is highly desirable in rotordyanmics to increase rigidty and increase the first critical speed.
The shaft couplers were also single piece, so the shaft couplers which do not have a edge true to the shaft axis. This meant that the shaft couplers could not be installed after the shafts were aligned, as the couplers would have to be installed before the shafts were aligned. The space the shaft couplers took up meant it was even more difficult to use the shaft stubs for alignment.
It was decided to use the flat faces of the torquemeter base as the reference for alignment. By designing the pump and turbine mounts to have a similar rectangular base and assuming that each shaft was true to the rectangular base, alignment was finally possible. Angular alignment was achieved by 3D printing a spacer with two parallel edges for

the pump or turbine to sit flat against onto the torquemeter. Then, radial misalignment was achieved by designing the face parallel to the shaft to be the parallel to the torquemeter face. A straight edge could then be used as a guide to align the shaft while tightening the mounting bolts. This procedure is shown in fig. 25.

Figure 25: Shaft alignment procedure using the flat faces of the component mounts and precision square blocks as references.

As the TM306 torquemeter's shaft with the base plate was machined to be preciscely 80.00 mm from the base plate, this same height was built into the pump and turbine mounts to achieve radial aligment in the vertical direction.

4.3 Water Circuit

Figure 26: P&ID of the water circuit. The tank feeds the supply pump, then the inlet throttle valve, the experimental pump, the outlet throttle valve and the return line. Pressure is measured at the supply (pt_{pump}), the pump inlet (pt_{in}) and the pump outlet (pt_{out}), and flow in the inlet line (fm0).

The water circuit is as follows: A water tank is pumped by the supply pump into an inlet throttle valve then into the experimental pump inlet. The pump outlet is then throttled by an outlet throttle valve before being returned to the water tank. Pressure sensors are placed at the inlet and outlet of the pump to measure the head rise across the pump. Flow rate is measured by a flow meter placed in the inlet line before the inlet throttle valve instead of the return line due to space constraints placed by the blast shield.

4.3.1 Inlet and Outlet Throttle Valves

The inlet throttle valve allows a reduction in pump inlet pressure to characterise the cavitation behaviour of the pump. The outlet throttle valve allows for a reduction in flow rate to characterise the head and cavitation behaviour of the pump at different flow rates.
The selection of the size of the inlet and outlet valves are crucial to the quality of the data. Ball valves were chosen as there were spares available, and they could be throttled in a controllable and repeatable manner by the angle of the valve handle. Custom 3D printed housings were designed to attach a servo motor to the valve handle in direct drive to allow for precise, remote and repeatable control of each valves.
Valve capacity is expressed throughout with the flow coefficient K_v, defined by

$$Q\,[\mathrm{m}^3/\mathrm{h}] = K_v\sqrt{\frac{\Delta p\,[\mathrm{bar}]}{SG}} \tag{56}$$

where SG is the specific gravity of the fluid, 1 for water. Measuring the flow rate and the pressure drop across a valve at a known opening angle therefore measures its K_v directly, which is how the characterisation below was done.
The main requirement for both valves were so that the range of required flow rates and pressure drop through each valve corresponded to a large range in valve opening angles. For the inlet valve, fully open must have a large margin to the cavitation limit of the pump, ideally, cavitation would start at around 50% valve opening and could be further reduced until the valve is closed. For the outlet valve, fully open must be at least 1.5 times the cut-off flow rate of the pump.
This requirement is further driven by the $1 - 3°$ resolution of the servo motors available in the department. Data for Kv against opening angle were extrapolated from online data from 1/8” in to 3/4” in were studied. However, as the data was extrapolated from data of ball valves with larger diameters, they had to be experimentally validated. The flow through a ball valve is also highly dependent on inlet and oulet conditions such as contraction or expansion of pipe diameter at the ports [?]. The size of my pipes were limited to what I could find in spare drawers, adding to the uncertainty.

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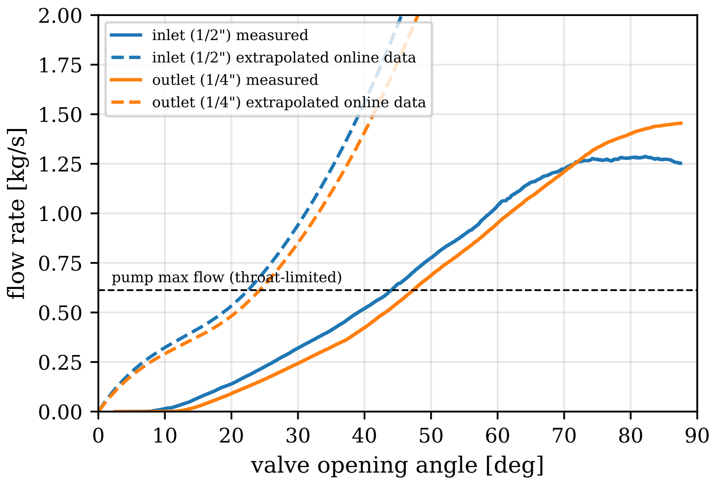
(a) Servo driven throttle valve assembly



(b) Extrapolated against measured valve capacity

Figure 27: Throttle valve characterisation. The extrapolated online ball valve data (dashed) overpredicts the measured flow capacity (solid) for both valves at their respective design pressure drops. The horizontal line is the throat limited maximum flow of the pump.

fig. 27b shows the experimentally measured Kv against opening angle values were lower than expected. However, this worked in my favour as it actually increased the range of useful valve opening angles.
The location chosen for the experiment required a safe test area and a control room separated to allow for the failure of the test rig without risk to the operator. For this reason, the experiment was conducted within the Imperial College Hypersonics Lab. The test room contains equipment that had to be protected, so a blast shield was designed and set up around the turbomachinery stand to protect the equipment from the possibility of mechanical failure and fragmentation.

4.4 Air supply

Originally, a 100 bar compressed air supply was designed and parts readied to be used to drive the turbine. Unfortunately due to time constraints, the shop air supply within the hypersonics lab was used instead. Details of the 100 bar compressed air supply can be found in . The shop air which was limited to 7 barg was connected to partially assembled parts of original feed system. The shop air was connected to a GRT20s-280N100-SSKN-L 0-280 to 0-100

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bar regulator with a Kv of 1.5. This regulator fed into the the main 1” valve, teed into a 1/2” vent valve and finally into the turbine inlet.
Both valves were Swagelok pneumatically actuated spring return ball valves. The main valve was the SS-65TS16-35C with a Cv of 40 and the vent valve was SS-45S8-33C with a Cv of 12. These valves were driven by 2 5/2 solenoid valves connected to the shop air supply.
Due to droop in the shop air regulator, maximum turbine inlet pressure at max flow was measured to be 3.1 bara. There was no budget for an air flowmeter.
The supply droop deserves a proper explanation because it directly sets the coupled test conditions in section 5. The shop air behaves as a regulated blowdown with a finite supply. When the turbine valve opens, the regulator cannot hold its set pressure at the demanded flow, so the stator pressure sags well below the static setting, and the achievable turbine inlet pressure under flow is roughly 38% of the design supply. This is a property of the rig, not of the turbine, but it has to be characterised because the turbine match point depends on it.
To verify the original 100 bar feed system before it was descoped, a transient flow network simulation was written from scratch. The model solves the tank blowdown as a system of differential equations, with each component represented by its flow coefficient, tube friction from roughness and bends, a regulator model holding its set pressure within a control band, and the choked stator as the terminating element. This was admittedly an overkill way to answer the question of whether two 3.8 litre tanks held enough gas for a full test, but it gave the test durations and the pressure transients at every node of the feed system, and the same component models were reused to size the valve Kv requirements. The simulation code is part of the project’s analysis repository.

4.5 Instrumentation and Data Acquisition

For cost and time reasons, a custom data acquisition and control system was designed to allow for remote operation of the test rig.

4.5.1 Main Architecture

Figure 28: Data acquisition architecture. Sensors and the DSP6001 analog outputs feed the MCP3208 ADC, read by an ESP32-S3 over SPI, streamed to the Raspberry Pi 5 over UART, logged to HDF5 and forwarded to the control room GUI over the network.

For safety reasons, the electronics were enclosed in a plastic box placed in the test room, controlled by a LAN cable fed under the wall between the test room and the control room into my laptop.
The main main computer inside this box was a Raspberry Pi 5 running a backend written in Python. The Raspberry Pi communicated with 3 ESP32S3 microcontrollers over a UART serial connection at 921600 baud. One ESP32S3 was used as the actuator controller, one was used as a ADC and flowmeter DAQ, and one was used as a thermocouple DAQ. Breadboards would require too many wires to maintain and unstable, while there was not enough time to make custom PCBs. Soldered perfboard was used to connect the components. The Raspberry Pi communicates through the LAN cable through UDP to a laptop in the control room, allowing for live data readout and control of the test rig with a GUI written in Python and Qt.

Figure 29: The control room GUI during a test: live channel readouts, strip charts of pressure, torque and speed, valve angle controls, and the test sequencing and logging controls. Every quantity shown is logged to HDF5 at the full 50 Hz loop rate, so the analysis of section 5 works from exactly what the operator saw.

4.5.2 Power Architecture and Custom Boards

The whole test cell runs from a single 24 V supply. This was a deliberate choice so there is only one thing to switch off and so the highest voltage anywhere in the cell stays low. Everything else is derived from the 24 V on the boards. A set of buck converters steps it down to the 5 V logic rail for the microcontrollers and to the minus 9 V and plus 9 V rails the torque op-amp needs. A separate LT3042 linear regulator then generates the quiet 5.11 V rail that acts as both the ADC reference and the pressure transducer excitation, kept off the switching converters so it stays clean. The reason that one rail feeds both the reference and the excitation is the ratiometric cancellation explained in the next section.

The electronics were not bought as modules, they were designed from scratch in KiCAD as a set of small boards that each do one job. There is the reference and ADC board carrying the LT3042 and the MCP3208, the torque conditioning board carrying the OPA2134 and its level shift, the current and voltage sensing for the motor with the WCS1600 hall sensor and a divider on the 24 V ESC supply, and the thermocouple front end built around the MAX31855 breakouts. Level shifting between the 3.3 V logic of the ESP32 and the 5 V SPI bus is done with a TXS0108E translator, with its weak internal pull ups disabled so the SPI edges stay clean. Because there was no time to have proper PCBs made, the boards were built on soldered perfboard, which is reliable enough as long as the layout keeps the noisy switching sections away from the quiet analog ones. The one real interference problem was the buck converters coupling magnetically into the signal lines, which was fixed by physically separating them from the analog boards and twisting the supply pairs.

4.5.3 ADC Measurements

Pressure, torque, RPM, voltage and current were measured with a MCP3208 8 channel 12-bit ADC connected to the ESP32S3. The MCP3208 was powered by a LT3042 LDO at 5.11V to reduce noise. This 5.11V was also used to power the pressure sensors. As the pressure transducers were ratiometric, both the ADC and the pressure transducers track bacck to the same voltage reference, reducing drift and noise.

The custom backend could only send UART signals back and forth to the ESPs at a 50 Hz loop. The MCP3208 was capable of sampling up to 100 ksps, so the ESP32 was programmed to collect 32 samples and average them per channel before sending the averaged data back to the Raspberry Pi. High frequency data couldn't be collected due to the UART connection being the bottleneck, but this averaging allowed for measurement of voltage at a resolution higher than 12-bits. For uncorrelated noise the averaging reduces the standard deviation of each logged point by

$$\sigma_{avg} = \frac{\sigma}{\sqrt{N}} = \frac{\sigma}{\sqrt{32}} \tag{57}$$

which is a factor of about 5.7, and is why the per bin scatter in the results is small.

The ratiometric arrangement deserves one line of algebra, because it deletes a systematic error term by design. The transducer output is proportional to its excitation voltage, $V_s = k\,p\,V_{exc}$, and the ADC reports the ratio of its input to its reference, $D = V_s/V_{ref}$. With both supplied from the same LT3042 rail, $V_{exc} = V_{ref}$, so

$$D = \frac{k\,p\,V_{exc}}{V_{ref}} = k\,p \tag{58}$$

and any drift of the shared rail cancels to first order, leaving only the transducer's own nonlinearity and the reference noise.
Originally, a separate computer running Magtrol's M-Test software was planned to be used to read the torque and RPM from the DSP6001. However, by connecting the DSP6001's analog output for torque and RPM to the MCP3208, the torque and RPM data could be integrated into the same DAQ system as the pressure sensors, allowing for time-exact data synchronisation and simpler data collection.
To measure the torque and RPM, the TM306 torquemeter was connected to the Magtrol DSP6001 dynamometer controller. The DSP6001 outputs a -10V to 10V analog signal for torque, and 0 to 10V for RPM. The DSP6001 was set to output 1V per 0.25 Nm and 1V per 5000 RPM. Because the MCP3208 could only measure positive voltages, an OPA2134 op-amp was used to shift the expected -5V to 5V torque signal to 0V to 5V. As the op-amp was not rail-to-rail, a buck converter was used to generate a -9V and 9V supply for the op-amp.
The motor's voltage and current were also measured by the ADC. The 24V supply to the motor ESC was measured with a voltage divider, and the current was measured with a WCS1600 DC hall effect current sensor.

Figure 30: ADC and signal conditioning schematic: the LT3042 reference powering both the MCP3208 and the ratiometric pressure transducers, and the op-amp level shift for the torque signal.

4.5.4 Other sensors

K-Type thermocouples although are less accurate for such a small temperature range were chosen for their cost. The thermocouples were read by an Adafruit MAX31855 thermocouple amplifier breakout board with integrated voltage regulator connected to the ESP32S3.
A E-WFS-GE turbine flowmeter with a range of 2 to 45 L/min with a cited accuracy of $\pm 10\%$ was used. To calibrate the flowmeter, a bucket and stopwatch was used to compare the flowmeter to the integrated flow rate over a period of time. The flowmeter was found to be accurate to $\pm 3\%$ [**?**]. The turbine generates pulses based on its rotation rate. The ESP32 measures the interval between these pulses to calculate the flow rate.

4.5.5 Actuator Control

One ESP32 is dedicated to actuation so the sensing loop is never held up by driving the actuators. It commands the two servos on the inlet and outlet throttle valves, holding each valve angle to the resolution the servo allows, and it drives the two solenoid valves that pilot the Swagelok pneumatic ball valves on the air side. All of this actuation power runs through the PILZ safety relay, so a trip or any loss of power drops every actuator to its safe state, the throttle servos going limp and the spring return air valves closing the supply and venting the manifold.

4.5.6 Cavitation Imaging

The transparent front casing and diffuser turn the pump into its own instrument, but only if the bubbles can actually be seen on camera. Cavitation bubbles are transparent; what a camera records is light scattered at the bubble interfaces, so the imaging arrangement matters as much as the casing. The casing was lit obliquely with a high intensity lamp placed off the camera axis, so that bubble interfaces scatter light toward the camera while the bulk water stays dark, and the polished resin surface was angled to keep the specular reflection of the lamp out of frame. Video was recorded through the polycarbonate shield with a consumer camera.

Two limitations of this arrangement should be stated. The frame rate is orders of magnitude below the blade passing frequency, so each frame is a time averaged record of the cavity envelope, not a snapshot of individual bubble dynamics, and the consumer camera's rolling shutter means different image rows sample different instants, geometrically distorting fast moving structures. Individual bubbles are still resolved in some frames where the motion is slow enough. These limits are acceptable for what the video is used for in section 5, identifying where cavitation occurs, when it onsets, and when the throat clears, all of which are envelope level questions. Bubble tracking or cavity dynamics would need a global shutter camera with a synchronised strobe, which is noted as future work.

Figure 31: The cavitation imaging arrangement: oblique illumination of the transparent casing with the camera viewing through the blast shield.

4.5.7 Instrument Uncertainty

table 5 lists every measurement channel with its range and systematic uncertainty. These values feed directly into the uncertainty analysis of section 5, where they are combined with the per point random scatter. The torque chain is the best characterised: the TM306 itself is rated to better than 0.1% of its 5 Nm range, the DSP6001 analog output adds 0.05%, and the conditioning chain was designed so the dominant systematic terms cancel, as described in the previous section. Where calibration certificates were not available, cheap cross checks bound the errors instead. The measured efficiency landing close to the design estimate validates the torque scale, the shutoff head scaling with speed squared through the origin bounds the pressure transducer gain error, and the repeatability of the non dimensional head coefficient across five speeds bounds the random error.

Table 5: Measurement channels, ranges and systematic uncertainties.

Quantity	Sensor	Range	Output	Uncertainty	Source
Pressure	TODO	TODO	TODO	TODO	TODO
Flow rate	TODO	TODO	TODO	TODO	TODO
Torque	TODO	TODO	TODO	TODO	TODO
Shaft speed	TODO	TODO	TODO	TODO	TODO
Temperature	TODO	TODO	TODO	TODO	TODO

4.6 Test Envelope and Limitations

The achievable test envelope was set by the drive hardware on the pump side and by the air supply on the turbine side. The electric motor had no speed control loop, only stepped throttle commands to the ESC, and the combination of the ESC current limit and motor torque cap meant pump testing topped out at around 7,500 RPM. On the turbine side, the shop air pressure and its droop under flow settled the coupled shaft at around 5400 RPM. Conditions above these speeds were not surveyed due to these experimental limitations, and the extrapolation of the measured behaviour towards the design point is handled analytically in section 5.

Safety drove three features of the test arrangement that are visible in the data taking: all testing was operated from a separate control room with nobody in the test cell, the rotating assembly was fully enclosed by a blast shield, and a hardware safety relay cuts all actuation power on an overspeed or emergency stop. The shield and the protection system were not bought off a shelf, they were sized by the calculations of the next section.

4.7 Containment and Protection System Design

Experimental turbomachinery has to be designed on the assumption that any rotating component or any non commercial pressure part can fail. The containment hardware was therefore sized from first principles. A note on the operating pressure: these calculations were performed for the original 100 bar stored air system that was later descoped to the 7 bar shop air supply, and the hardware they sized was retained, so the as-tested system carries safety factors far beyond what the 7 bar regime requires.

4.7.1 Fragment Energy and Blast Shield Sizing

For the failure of a pressure holding component, the fragment velocity is estimated with the method of Baker [41], developed from simulations of a vessel breaking into sectional fragments and matched to experiment within 10%. The non-dimensional burst pressure is

$$\bar{P} = \frac{(p_0 - p_e)\,\mathcal{V}}{m_c\,\gamma R T_0} \tag{59}$$

where p_0 is the vessel pressure at failure (the full 100 bar, assuming both the regulator and the burst disc have failed), p_e is atmospheric pressure, $\mathcal{V}$ the vessel volume and m_c the casing mass. The worst case line of Baker's velocity chart is

$$\log_{10}\bar{u} = 0.6\log_{10}\bar{P} + 0.15, \qquad \bar{u} = \frac{u}{\sqrt{K\gamma R T_0}} \tag{60}$$

with $K = 1$ for evenly distributed fragment mass. The only non commercial pressure holding component is the stator, and this procedure gives a fragment velocity of 72 m/s. Assuming the stator splits into two 0.1 kg fragments,

$$E = \tfrac{1}{2}mu^2 = \tfrac{1}{2}\times 0.1\times 72^2 = 261\ \text{J} \tag{61}$$

For the rotating assembly, the bounding energy is the full rotational kinetic energy of the blisk at design speed,

$$E = \tfrac{1}{2}J\omega^2 = \tfrac{1}{2}\times 6\times 10^{-5}\times 2094^2 = 131\ \text{J} \tag{62}$$

The main line of defence is a 20 mm Dyneema (UHMWPE composite) ballistic panel on an aluminium extrusion frame. Its capacity is expressed as a specific energy absorption,

$$SEA = \frac{E}{m}, \qquad E_{max} = SEA\cdot A\,t\,\rho \tag{63}$$

where A is the projected area of the impacting fragment, t the panel thickness and ρ the panel density. The SEA is derived from published v_{50} ballistic limits for fragment simulating projectiles against 20 mm panels [42]: a 12.7 mm FSP gives 1.84 MJ/kg and a 20 mm FSP gives 1.68 MJ/kg, the latter agreeing with independent flat projectile data. Using the conservative value, the panel can absorb 163 kJ over the stator's projected area and 32.6 kJ over the blisk's, against the fragment energies of eqs. (61) and (62). The containment margins are roughly 600 and 250.

A 15 mm polycarbonate shield additionally separates the turbomachinery from the dynamometer. Its capacity is estimated with the shear plug model [42], where the projectile shears a plug of perimeter s_p out of the sheet:

$$E = \int_0^t s_p\,\tau_{pc}\,(t-x)\,dx = \tfrac{1}{2}s_p\,\tau_{pc}\,t^2 \tag{64}$$

with $\tau_{pc} = 60$ MPa for polycarbonate, conservative because the material strain hardens at impact rates. This gives 2.7 kJ for the stator perimeter and 4.1 kJ for the blisk perimeter, safety factors of 10 and 30 on their own.

4.7.2 Bounding Shaft Torque

The rig torque rating was checked against an upper bound from the Euler equation, taking the maximum mass flow, full tangential jet velocity at entry and perfect counter rotation at exit:

$$\tau_{max} = \dot{m}\, r\, (c_{u,in} - c_{u,out}) = 0.034 \times 0.05 \times (469 + 469) = 1.6 \text{ Nm} \tag{65}$$

This is an unachievable worst case and still sits well below the 3 Nm rating of the shaft train.

4.7.3 Protection System Electronics

The protection logic that the next section's transient analysis relies on is implemented in hardware, not software. The core is a PILZ PNOZ X3 safety relay wired as a dual channel emergency stop chain: the control room E-stop button and the overspeed trip both open the chain, and the relay then cuts the supply to every actuator solenoid through its force guided contacts. The safety relay is a certified device whose failure modes are themselves controlled, which is the reason for using one instead of a software interlock through the Raspberry Pi: the software stack is involved in detecting the overspeed, but once the trip fires, no software is in the loop between the relay and the valves. The pneumatic actuators are spring return, chosen so that the de-energised state is the safe state: on any loss of actuation power, including a tripped relay, a cut LAN cable or a power failure, the main air valve drives closed and the vent valve opens the manifold to atmosphere. The 20 ms figure used in the runaway analysis below is the relay contact response; the valve travel times dominate the rest of the shutdown timeline.
The electrical system itself is low risk by construction: everything in the test cell runs at 24 V DC or below from a single supply, the electronics live in an enclosed plastic box protecting them from spray, and the only conductor crossing the test cell boundary is the LAN cable to the control room, which carries no power. The motor ESC and its 24 V supply are the highest energy electrical components, and they are inside the same enclosure boundary with the rotating hardware, behind the shield.

4.7.4 Overspeed Runaway Transient

The protection system has to act faster than the worst credible overspeed. With a broken shaft leaving only the turbine, blisk and bearings, the angular acceleration under the maximum torque at the 17,000 RPM redline is

$$\dot{\omega} = \frac{\tau}{J} = \frac{0.9}{2.74 \times 10^{-4}} = 3285 \text{ rad/s}^2 \approx 31{,}300 \text{ RPM/s} \tag{66}$$

After the trip, the manifold between the closed valve and the stator continues to blow down through the turbine. In choked flow the mass flow is proportional to the remaining manifold mass, an exponential decay with time constant

$$\frac{dm}{dt} = -\dot{m}_0 \frac{m}{m_0} \quad \Rightarrow \quad \tau_{bd} = \frac{m_0}{\dot{m}_0} = 42 \text{ ms} \tag{67}$$

for the measured manifold volume. The protection sequence is: overspeed detected at $t = 0$, all actuation power cut by the hardware relay at 20 ms, main valve fully closed at 170 ms, manifold vented by around 200 ms. Holding the maximum torque over the full 200 ms overestimates the excursion and still bounds it at 6,260 RPM, so an overspeed tripped at 17,000 RPM peaks below 23,300 RPM, briefly exceeding the 20,000 RPM dynamometer rating within the manufacturer's stated tolerance for transient overuse. Cavitation testing was further speed limited so that the same worst case stays below 19,000 RPM. The venting noise and reaction force calculations that complete the safety case are in section B.

5 Results and Discussion

5.1 Experimental Campaign

The experimental campaign was conducted in three phases: commissioning, electric motor driven pump testing, then integrated turbine and pump testing. The standalone turbine testing was not conducted due to time constraints. This means the turbine is assessed through its models and through the coupled test, and a standalone brake characterisation is the first item of future work.
Within each run the shaft speed was not perfectly constant, so the data points were adjusted to the run median speed using the affinity laws before binning. The adjustment is small, validated to around 0.2% on the head coefficient, and it is what allows each run to be treated as a single speed curve.
All error bars in this chapter follow the same simple scheme. The systematic uncertainty of each instrument comes from table 5, the random uncertainty is the standard error of the median within each bin, and the two are combined in quadrature at roughly 95% confidence, $U_Y = \sqrt{B_Y^2 + (2P_Y)^2}$. Propagation into derived quantities like efficiency and the non dimensional coefficients is done by linear propagation. One design decision from section 4 matters here:

the low head might due to leading edge r ing as by lock maybe due to rotatio the eq tions s gest.

look at H-Q d and se the ca itation

because the pressure transducers and the ADC share the same reference voltage, excitation and gain drift cancel to first order and do not appear as a systematic term. Where error bars are not visible they are smaller than the markers.

5.2 Pump: Head–Flow Characteristics

There are two main curves that are used to evaluate pump performance: the head-flow curve and the efficiency curve. The head-flow curve shows the achievable head at different flow rates and will be discussed first.

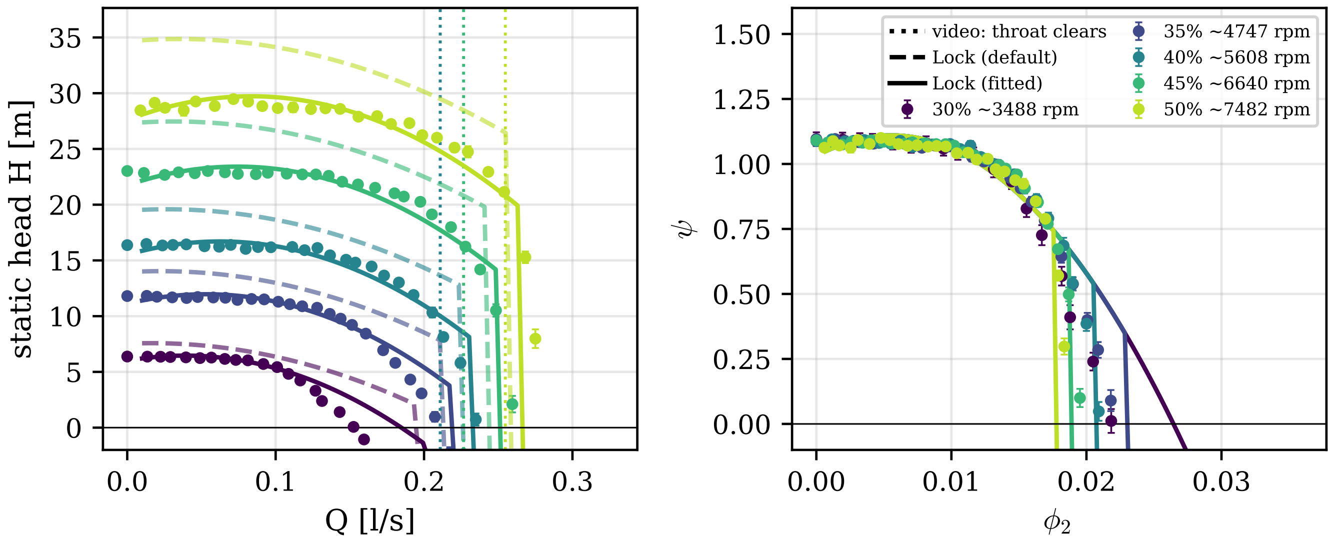

Figure 32: Measured head against flow rate across five speeds (left) and the same data collapsed to head and flow coefficients (right). Dashed lines are Lock's model with literature constants, solid lines are the bounded re-fit with the as-built throat. Vertical dashed lines mark where the synchronised video shows the diffuser throat cavitation clearing. Error bars are systematic and random combined in quadrature at 95% confidence, smaller than markers where not visible.

fig. 32 Shows the actual achieved head rise across flow rates at the different RPMs that were tested. Overlayed are solid and dashed lines. Both are theoretical head-flow curves generated from eq. (24). The dashed line is with exactly the parameters from Lock's paper, and the solid line is with the pre-rotation K_1 increased from 0.17 to 0.53. The dashed vertical dashed line shows the point at a synchronized video camera detected cavitation within the diffuser through the transparent diffuser wall. It can be seen that the original parameters overpredict the head rise, and there are many mechanisms that could be causing this.

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The non dimensional view in the right panel carries the headline numbers. The five speeds collapse onto a single curve at a head coefficient of $\psi = 1.10 \pm 0.05$ in the operating region, against a design value of 1.065, so the pump delivers its design head with around 3% to spare. The collapse itself is the experimental proof of dynamic similarity, and it is what licenses the affinity adjustment within runs and the extrapolation to higher speeds later in this chapter. The diverging tails at high flow coefficient are a rig artifact rather than a breakdown of similarity, because the fixed feed pressure means the maximum achievable flow coefficient falls as speed rises.

The fitted constants need to be defended rather than just stated, because a fitted model proves nothing if the constants are unphysical. The bounded re-fit with the as-built throat diameter fixed at 3.0 mm gives $K_1 = 0.53$ and a diffuser loss coefficient of 0.42, both interior to their physical bounds. The excess of K_1 over Lock's own 0.15 to 0.20 study has named candidates that are all in Lock's paper: the deliberate prerotation designed in through $d_1 = 1.1d_0$, the rounding of the printed blade leading edges which Lock notes reduces effective head, and the optimism of his effective radius assumption. The diffuser loss of 0.42 against the clean pipe entry value of 0.194 is consistent with the entry separation mechanism Lock describes at the diffuser mouth. Alternative explanations can be ruled out: an error in the torque or pressure scales would shift the dimensional curves but not change the shape of the droop, and a slip type explanation would move the intercept of the non dimensional curve rather than its slope.

The most important feature of fig. 32 is the high flow cutoff, and here the transparent casing converts a fit into a validation. On every head flow ramp the video shows cavitation in the diffuser throat clearing as the flow falls, frame matched to the data at three flow rates across three runs. With the true throat diameter of 3.0 mm in the model, Lock's vapour pressure cutoff lands on all three video clearing points to within roughly 10%, and on the measured runout. The history here is worth stating. With the nominal cutwater diameter of 3.8 mm the model overpredicted the cutoff badly, and it was the systematic disagreement with the video that prompted the CAD section check which found the true minimum area. The model diagnosed a wrong geometry input, which is a stronger validation than agreement on the first try. The remaining discrepancies are also informative. The measured cutoffs sit 5 to 10% later than predicted, consistent with a small residual flow contraction, and the head begins rolling off slightly before the cavitation clears, which cavitation cannot explain and which points to the same diffuser entry separation responsible

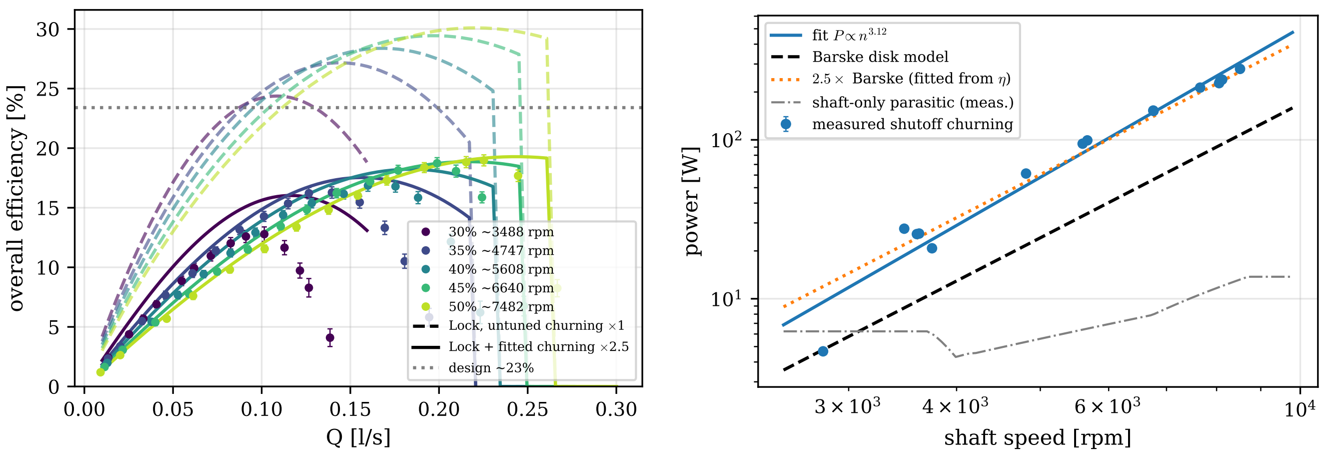
for the elevated loss coefficient. Collector losses growing with flow at low specific speed is independently reported, as Khan's volute head loss curves blow up at overload in exactly this way [43].
The conclusion of this section is then a mechanism, not a fit: the maximum flow of this pump is limited by cavitation in the diffuser throat, established independently by the model and the video.

5.3 Pump: Efficiency and Churning

The overall efficiency is computed from directly measured quantities only,

$$\eta = \frac{P_{hyd}}{P_{shaft}} = \frac{\rho g Q H}{\tau \omega} \tag{68}$$

with the torque measured at the shaft between the drive and the pump, so the drive losses never enter.
The efficiency curve generated by measuring shaft power, subtracting mechanical losses can be shown in the following graph:
The right graph shows the predicted efficiency curve from pure theory, and the right graph shows the curve but with adjusted disc friction. To explain the disc friction adjustment, the pump data at zero flow where all the shaft power was only going into hydraulic losses:



(a) Overall efficiency against flow rate

(b) Shutoff churning power against speed

Figure 33: Pump efficiency and the churning measurement. In (a) the dashed curves use Barske's textbook disc friction and overpredict the efficiency, while the solid curves apply the measured churning multiplier. In (b) the churning power is measured directly at shutoff, where shaft power minus the measured shaft-only parasitic is all churning; the full campaign of 15 shutoff points is used, not just the five canonical runs. Error bars as in fig. 32.

fig. 33b shows that the measured disc friction losses are about 2.5 times higher than barske's formula for disc friction. Adjusting the disc friction by this factor leads to a much better fit of the efficiency curve, as can be seen in fig. 33a.
The way this multiplier was measured matters, because it changes its status from a tuning factor to a measurement. At shutoff the pump moves no flow, so after subtracting the measured shaft-only parasitic power, everything the shaft delivers goes into churning the flooded casing. Across 15 shutoff points from 2,800 to 8,500 RPM the measured churning sits at 2.5 to 2.9 times the Barske disc correlation, with a mean of 2.54, and follows a power law of $n^{3.1}$, between Barske's exponent of 2.8 and the cubic forms in Gülich. The factor of 2.5 used in the efficiency model was originally fitted to the efficiency curves, so its agreement with the shutoff measurement is corroboration by data that did not produce it. The points below about 4,800 RPM scatter heavily and were excluded, as the torque there sits at the noise floor of the measurement chain.
The peak measured efficiency is $(19 \pm 2)\,\%$ against a design estimate of 23.4 %. Where the power actually goes at the best efficiency point is summarised in fig. 34. Churning dominates everything: at the 7,500 RPM best efficiency point roughly two thirds of the shaft power goes into churning the flooded casing, mechanical losses take a few percent, the internal flow losses take around a tenth, and only about a fifth of the shaft power leaves as useful hydraulic power. The attribution to the open impeller architecture is the point, as the whole casing is flooded and churned while only one sector does useful pumping. This is the price of the Barske architecture at low Reynolds number, not a defect of this particular build.

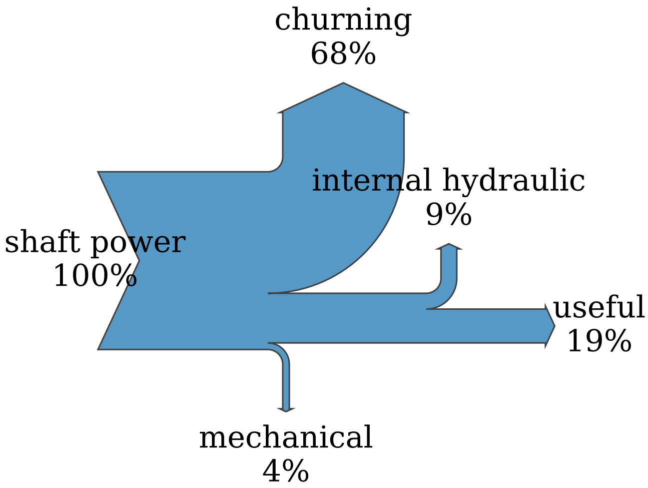


Figure 34: Shaft power budget at the measured best efficiency point of the 7,500 RPM run. Useful hydraulic power is measured, churning uses the shutoff-measured multiplier at this speed, mechanical losses are the measured shaft-only parasitic, and the remainder is internal hydraulic loss.

The natural question is what this efficiency becomes at the design speed, and the answer comes from the data rather than an assumption. Each run sits at a different Reynolds number, so the five runs themselves measure how the losses scale. fig. 35 plots the peak efficiency of each run against speed and fits the loss ratio model $(1-\eta) \propto Re^{-a}$, which is the standard form for friction dominated loss scaling. The fitted exponent is $a = 0.098 \pm 0.010$, essentially the textbook Moody value of 0.1, and extrapolating along the fit gives an efficiency of 26.4% at 20 000 rpm with a band of 25.7 to 27.1%. The exponent is measured from this machine, not assumed, which is the difference between an extrapolation and a guess. There is also precedent for efficiency recovering with Reynolds number in exactly this class of machine, as a NASA ten times scale test of a low specific speed rocket pump took the efficiency from 58.7% to 68.9% on Reynolds number alone [8].

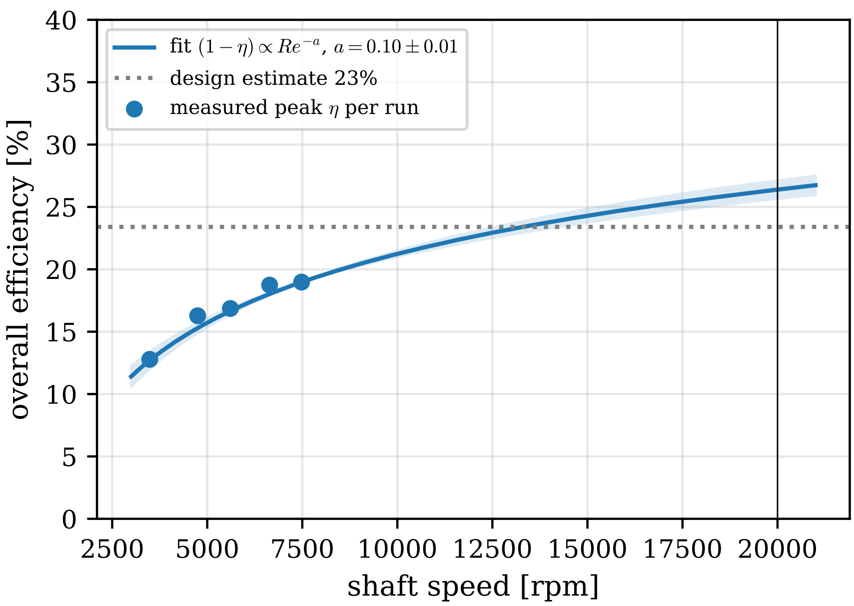


Figure 35: Peak overall efficiency per run against shaft speed, with the fitted loss ratio $(1-\eta) \propto Re^{-a}$ and its uncertainty band extrapolated to the design speed. The fitted exponent $a = 0.098 \pm 0.010$ matches the textbook friction scaling.

For context on whether around 19% measured and around 26% extrapolated is good or bad, the conventional counterfactual exists at literally this specific speed. Kim's closed impeller design at the same n_q reaches 67% in CFD, but it is unvalidated and requires 0.31 mm closed channels, the geometry this project's own conventional design attempt showed to be unmanufacturable [6]. The CFD studies of Barske type machines report higher efficiencies than measured here at higher Reynolds numbers [5, 7], while the historic measured partial emission pumps sit at 35 to 60% at much larger scale. The gap between this machine and those numbers is accounted for by the measured churning and the

sub design Reynolds number, not by an unexplained deficiency.
One tension should be stated openly because it bounds the extrapolation. Carrying the churning multiplier of 2.5 unchanged to 20 000 rpm would predict around 3 kW of churning alone, more than the entire turbine power budget. The design efficiency therefore requires the multiplier to fall with Reynolds number, which is the same physics the fitted exponent already encodes. The multiplier is measured at test Reynolds numbers, and its Reynolds dependence is the explicit assumption of the extrapolation.

5.4 Pump: Cavitation

Specifically for rocket engine applications, the cavitation behaviour is of high importance. The following graph shows the cavitation curve generated by plotting the head at which cavitation was observed at different flow rates:

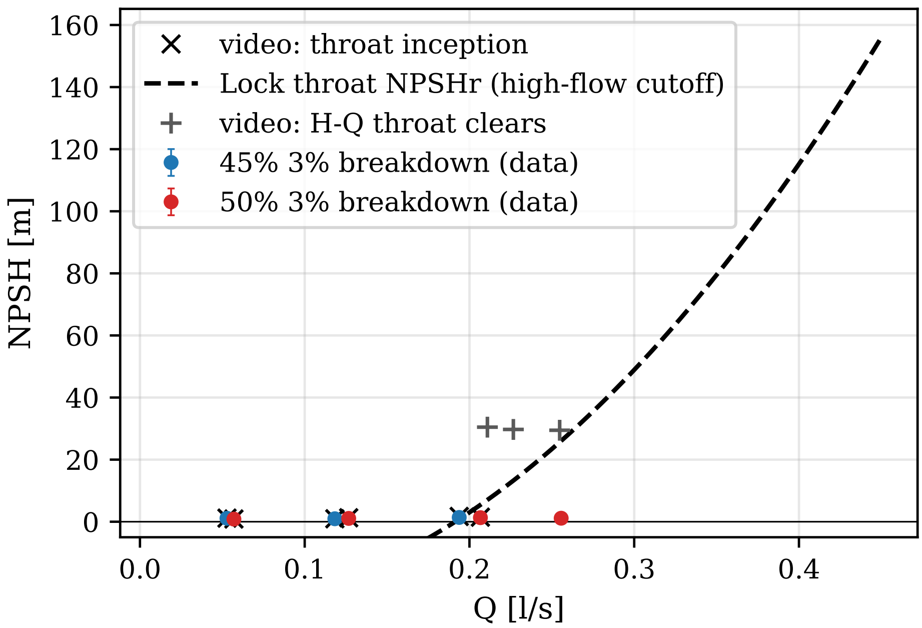


Figure 36: NPSH at 3% head breakdown against flow rate, with the visually observed throat cavitation onsets and the Lock throat cutoff curve. Error bars as in fig. 32.

fig. 36 shows the NPSH required to avoid cavitation at different flow rates. The dashed line is the theoretical curve generated by solving eq. (26) for the flow rate at which the head at the diffuser throat is equal to the vapor pressure head. Two interesting observations here.
The full suction test behaviour is shown in fig. 37 in the standard form, head normalised by the fully wetted reference against NPSH available, one curve per held flow rate, the same presentation Gülich uses for suction tests at constant speed. The curves hold their reference head over most of the suction range and then drop over a cliff. The 3% criterion line picks the breakdown point off each curve, and the cliff is steep enough that the choice of criterion barely moves the answer, as the head loses around 60% over a range comparable to the binning noise.

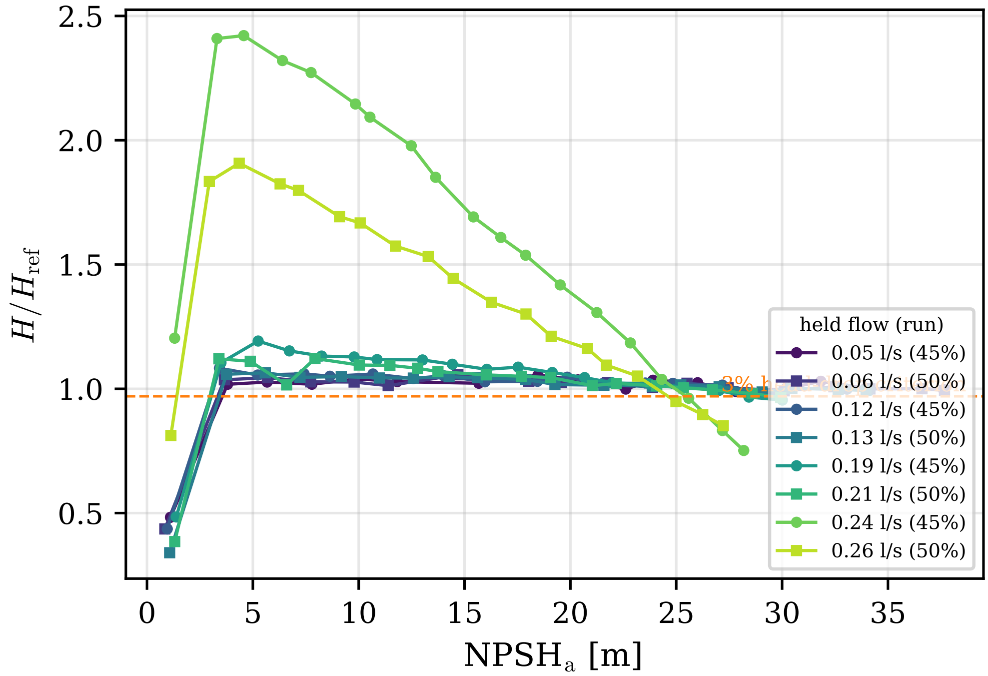


Figure 37: Suction test curves at constant speed: normalised head against NPSH available, one curve per held flow rate, by successive reduction of the inlet pressure. The horizontal line is the 3% head-drop criterion used for fig. 36.

The result that makes this section more than a standard suction test is where the breakdown happens. Two independent methods agree: the 3% breakdown from the pressure data coincides with the visually observed throat cavitation to around 0.1 m of NPSH, while cavitation at the impeller eye appears far earlier and does nothing measurable to the head. The naive reading of pump cavitation, bubbles at the inlet eye killing the head, is simply wrong for this machine. The eye cavitates benignly, and it is the diffuser throat that drives breakdown. This is Lock's diffuser limited claim, confirmed and this time directly visualised. One caveat applies to the measurement method: the inlet throttle drops the flow as well as the suction pressure, so the rigorous breakdown metric is the cliff knee rather than a fixed reference percentage, and the difference is small here because the cliff is steep.

Packaged as a single literature comparable number, the suction specific speed at breakdown is around 75 in metric form. That is close to the cavitation onset value of around 64 reported for a Barske type impeller in the same regime [5], and far below the industrial figure of around 200, because in this architecture the throat sets breakdown rather than the eye.

The throat sizing itself also lands in an interesting place against the literature. The throat to tip diameter ratio of this pump, about 0.051, follows the classical Barske and Balje guidance, the same starting point Danieli used for his partial emission rocket pump, whose initial throat ratio of 0.064 failed his head requirement. His CFD optimisation recovered head and efficiency by opening the throat ratio to 0.18 to 0.28 [7]. That independently corroborates the finding here that the classical throat under delivers and limits the runout, and it hands the design a cited future work item: enlarge the throat, traded against the suction margin that partial emission buys. His flow rate is 2.7 times larger and his study is CFD only, so it is the trend that is being cited, not the numbers.

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(a) Fully wetted

(b) Eye cavitation only

(c) Throat cavitation

(d) Both sites, near breakdown

Figure 38: Stills through the transparent front casing at descending NPSH. The eye cavitates first with no measurable head effect, while the appearance of throat cavitation coincides with the head breakdown. Frame timestamps are synchronised to the logged data.

5.5 Coupled Turbopump

The integrated test is the proof of concept of the whole project. With the turbine driving the pump on shop air, the shaft settled at around 5400 RPM, the pump produced a measurable head rise, and the assembly survived repeated runs. The settling itself is a result: the speed approached the match point monotonically and held it without any active control, which is the self regulating behaviour predicted by the torque matching argument of section 2. The operating point is deep off design, with the stator pressure at roughly 38% of the design supply and a blade to jet speed ratio of around 0.06 against the design 0.2. fig. 39 shows the coupled pump data landing on the motor driven curves, which is the cross validation between the two test configurations.

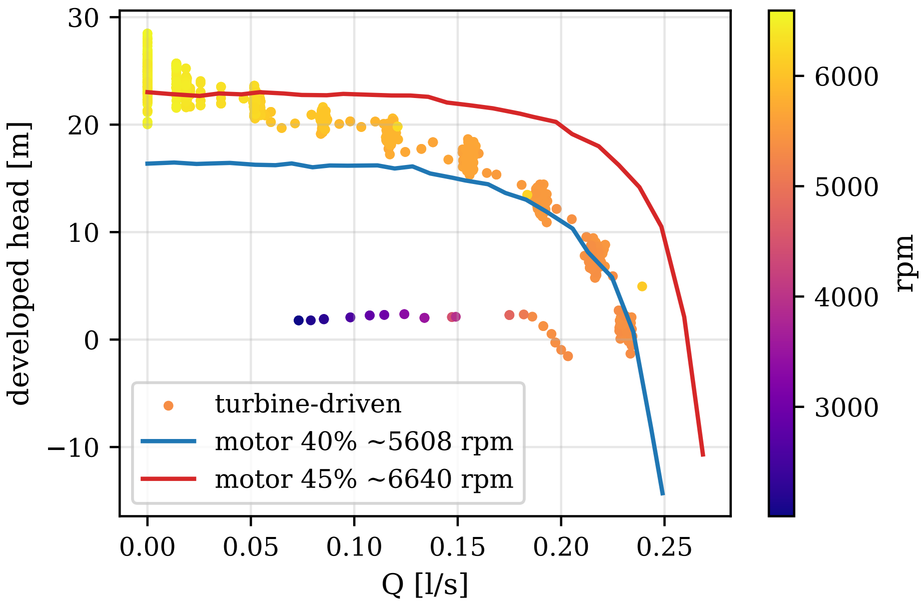

Figure 39: Coupled turbine driven pump operation against the motor driven characteristic curves. The coupled data, coloured by shaft speed, lands on the same head flow behaviour measured under motor drive.

The turbine side needs more care, because the rig has no air mass flow meter. Without a measured mass flow, a single forward efficiency number for the turbine would be under determined: a choked mass flow with a weak jet and a reduced mass flow with a near design jet both fit the same measured torque. Quoting one number would be misleading, so this thesis does not.

The torque models compared against the data in fig. 40 are built from the Euler equation with the choked mass flow of eq. (45), which remains valid regardless of the rotor state. For a rotor receiving an effective tangential jet velocity c^e_{3u}, the shaft torque is

$$\tau = \dot{m}\,(1 + k_R)\,r_m\,(c^e_{3u} - u) \tag{69}$$

where the factor $(1 + k_R)$ carries the contribution of the turned relative flow at exit. The started model sets c^e_{3u} from the full design expansion at the measured supply pressure, while the unstarted model treats the effective jet as a fraction of it, fitted to the data.

What breaks the degeneracy is the starting analysis from section 3. At the test shaft speed the blade speed is so low that the rotor inlet relative Mach number rises to 1.58, above the Goldman starting limit of 1.408, so the rotor passages cannot swallow the startup shock system and the rotor runs unstarted. The design stage prediction was that the rotor cannot start below approximately 15 300 RPM, and the coupled test never exceeded half of that. An unstarted rotor sits behind a detached bow shock, the passage flow is subsonic, and the jet momentum reaching the blades collapses. fig. 40 shows the consequence: the measured torque of around 0.27 N m sits a factor of four below the started flow model at around 0.87 N m, and an unstarted model with an effective jet of around a third of the design velocity passes through the data. The near design jet branch of the degeneracy is ruled out precisely because it would require a started rotor, which the starting criterion forbids at this speed. The torque deficit is therefore not an anomaly, it is a diagnosed off design state that the design stage analysis predicted before the test was run.

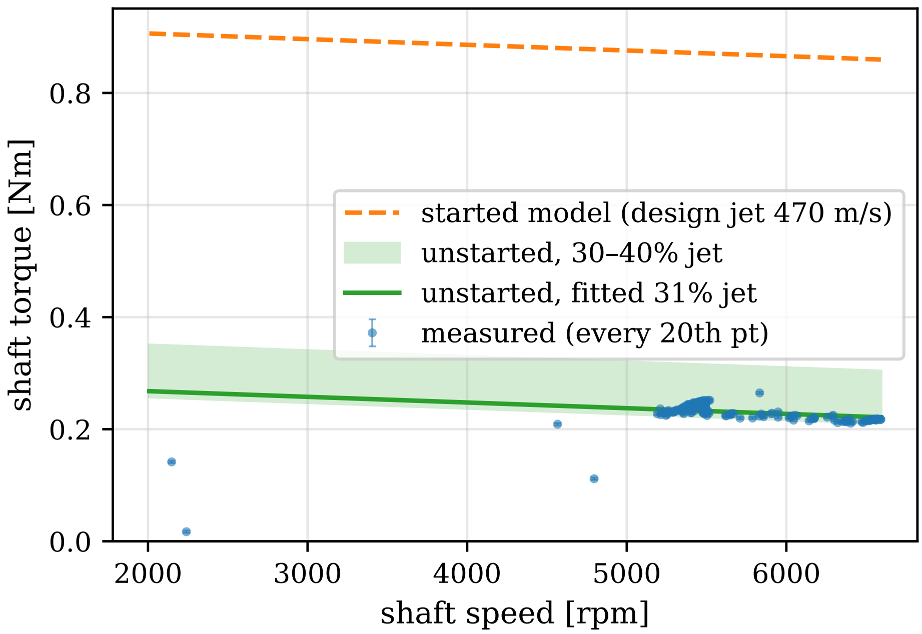

Figure 40: Coupled shaft torque against speed: measured data, the started flow model (dashed) and the unstarted model with a reduced effective jet (solid with band). The factor of four deficit against the started model is explained by the rotor running unstarted below the starting threshold.

Two rigour checks defend this diagnosis. First, the nozzle can be ruled out as the location of the loss. At the test supply pressure the over expanded conical nozzle sits at a pressure ratio of about 0.61 to ambient, above the separation threshold of roughly 0.4 from the rocket nozzle separation criteria [44], so the nozzle remains attached and supersonic and the velocity collapse must happen at the rotor. Second, the threshold itself should not be over read. With three discrete nozzles the rotor relative Mach number varies around the admitted arc, so the 15 300 RPM figure is a mean flow estimate and operation near it may start and unstart intermittently [11].

Because the stage was never measured in its started state, the design point efficiency prediction has to lean on external evidence, and the literature provides a tight bracket. The closest comparison is a machine of the same architecture, a radial inflow supersonic partial admission impulse stage at Mach 1.65, where the measured debit from ideal efficiency was 28 percentage points [39]. The Weiss chain here predicts 65% falling to 0.47, a debit of just under 28 points, so the loss model magnitude is corroborated experimentally to about one percentage point on the same machine class, with the caveat that his rotor ran started. Peng's 1.5 kW partial admission supersonic stage gives a CFD efficiency of around 42.5% [13], bracketing the prediction from below. NASA tested a Mach 2 stage from full admission down to 12.5% admission and found the full admission efficiency at 42.5% with the drop to 12.5% admission costing only around 5 points [45], which matters here because it acquits partial admission: at $\varepsilon = 0.15$ the admission penalty is modest, and the measured deficit belongs to the unstart, not the admission. Ohlsson's closed form partial admission theory, inside whose validation range this turbine's arc to chord ratio sits, composes with his low aspect ratio correlations to a literature only stage efficiency estimate of around 0.60 at his design speed ratio [23, 22], a second independent bracket. Finally, a precedent exists for exactly this failure mode: the SLA resin turbine Weiss cites also maxed out below its design speed with the cause left unexplained [20]. This work supplies the diagnosis that precedent lacked, the combination of supply droop and rotor unstart.

5.6 Extrapolation to the Design Point

The design point of 20 000 rpm was not reached in testing, so every claim about it is an extrapolation, and the value of this section is being precise about which extrapolations are licensed and which are not. On the pump side the licence is the non dimensional collapse of fig. 32. Because five speeds collapse onto one head coefficient curve, the affinity laws demonstrably hold for this machine, and fig. 41 carries the pooled curve to the design speed, where it passes through the design point. The efficiency extrapolation rides on the fitted Reynolds exponent of fig. 35 and arrives at 26.4% with a stated band.

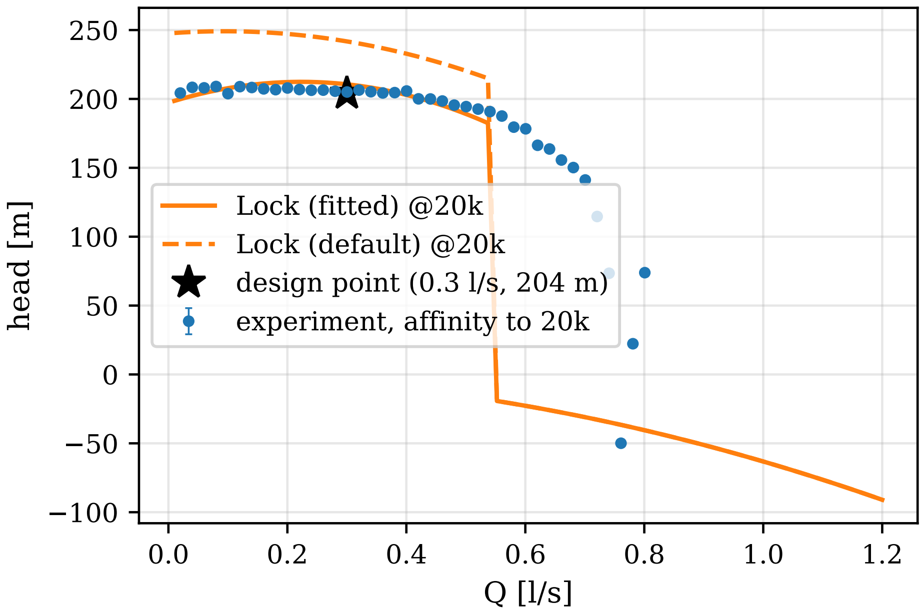

Figure 41: The pooled non dimensional characteristic carried to 20 000 rpm by the affinity laws, with the Lock model at design speed and the design point marked.

On the turbine side the claim has to be narrower. The coupled test ran at around 5400 RPM with the rotor unstarted, so the stage similitude parameters at design, the rotor relative Mach number and the blade to jet speed ratio, were never matched. What is validated are the speed independent pieces: the choked nozzle, the MoC profile generation against Goldman's published case, and the starting criterion itself, whose off design prediction the coupled test confirmed. Two named gaps remain on the road to 20,000 RPM. First, the started stage efficiency of 0.47 is a corroborated prediction, not a measurement, and stays that way until a test above the starting threshold. Second, the working fluid: cold air at $\gamma = 1.4$ stands in for hot gas at around 1.2, which shifts the expansion and the starting margin.

5.7 Synthesis

table 6 collects the quantitative comparisons of this chapter in one place.

Table 6: Predicted against measured summary. Percentage differences are computed by the analysis pipeline from the same registry that supplies every number in this thesis.

Quantity	Predicted	Measured	Δ	Note
head coefficient ψ	1.065	1.10 ± 0.05	+3 %	Barske design vs measured (5-speed c
overall efficiency η at 20 000 rpm	23.4 %	27 %	+16 %	design estimate vs Re-fit extrapolatio
shutoff churning power	Barske disk correlation	2.5 ×	+150 %	measured at shutoff, 15 points
rotor-inlet relative Mach M_{w3} at test	1.408	1.58	+12 %	vs Goldman starting limit → unstarte
coupled shaft torque	0.87 N m	0.27 N m	−69 %	started-flow model vs measured (unst

On the adequacy of the toolchain, the verdict is positive within stated limits. Lock's model reproduces the head curve including its droop, and places the cavitation limited runout at the right flow once the as-built throat is used, so it is adequate for design stage screening of this machine class. The Barske sizing model delivered a pump that meets its design head coefficient within 3%. On the turbine, the speed independent codes validated cleanly, and the loss chain prediction is corroborated externally to about a point, but the started stage remains unverified by this rig. The 1D and analytical toolchain was sufficient to design, predict and diagnose everything this project measured, and nothing in the data demanded CFD.

On the Barske architecture itself, the central question of the thesis, the data supports a measured verdict. The architecture does what it promises at this specific speed: it is manufacturable where the conventional impeller is not, it delivers its design head, and its suction behaviour is dominated by a single identifiable mechanism that a transparent casing can watch. The price is efficiency, with churning measured at two and a half times the textbook estimate eating two thirds of the shaft power at test speed. Whether around a quarter efficiency at design speed is acceptable depends entirely on the application, and for expendable small rocket stages where feed pressure mass dominates, it plausibly is.

One mechanical learning point belongs in the record. The first rig iteration used printed plastic load bearing shafts, which crept and failed under high speed loads, and the redesign moved every rotating load path to metal, a stainless pump shaft and an aluminium turbine shaft. Printed structural parts are acceptable for casings and blading at these loads, but not for shafts.

6 Conclusions

A small scale low specific speed turbopump, a Barske pump driven by a supersonic partial admission impulse turbine, was designed, built and tested, and its measured behaviour was used to answer the three objectives set in section 1.4.
Objective 1, pump characterisation. The Barske pump delivers its design head, with the measured head coefficient of 1.10 ± 0.05 sitting 3% above the design value and collapsing across five speeds, which demonstrates dynamic similarity. The peak measured efficiency is (19 ± 2) %, and the dominant loss was identified and measured rather than assumed: churning of the flooded casing absorbs about two thirds of the shaft power at test speed, at 2.5 to 2.9 times the textbook disc friction estimate, measured directly at shutoff. The cavitation behaviour was characterised and, through the transparent casing, visualised: the impeller eye cavitates early and benignly, while breakdown is driven by the diffuser throat, with the 3% breakdown and the visual throat onset agreeing to around 0.1 m. Against the design methodology, Lock's model reproduces the measured head droop and locates the cavitation limited runout once the as-built throat geometry is used, the affinity laws hold demonstrably, and the fitted Reynolds scaling carries the efficiency to 26.4% at the design speed with a stated band.
Objective 2, turbine design and model validation. The turbine was designed with a toolchain of MoC blade generation, the Sasman and Cresci separation check, the Goldman starting criterion with Colclough's empirical derating, and the Weiss loss chain, giving a predicted total to static efficiency of 0.47 against an ideal 65%. The standalone brake test was not performed, so the started stage efficiency remains a prediction, corroborated to about one percentage point by a measured machine of the same architecture. The starting criterion proved to be the load bearing piece of the toolchain, as its off design prediction explained the coupled test result.
Objective 3, integration. Coupled turbopump operation was demonstrated at around 5400 RPM, with the turbine driving the pump to a measurable head rise and the shaft settling at a stable self regulating match point with no active control. The gap to the design speed is explained by two named and quantified mechanisms, the supply pressure droop to roughly 38% of the design stator pressure and the predicted rotor unstart below approximately 15 300 RPM, rather than by any unexplained deficiency of the machine. Taken together, the coupled result shows the combined 1D and analytical toolchain is sufficient for design stage prediction of this class of machine, and no result in this work required CFD to predict or to explain.
Overall, this work provides an open, experimentally validated characterisation of a Barske turbopump at $n_q \approx 6.4$, including a measured loss breakdown and a direct visualisation of the cavitation mechanism that sets its operating limit.

6.1 Future Work

Each item below names the uncertainty it resolves.

1. Standalone turbine characterisation against the hysteresis brake, above the starting threshold of approximately 15 300 RPM. The configuration is built but was never run, and it resolves the single largest open prediction, the started stage efficiency of 0.47.
2. A 15 mm direct rigid coupler to unlock the full 20,000 RPM envelope of the torquemeter, removing the drive side speed limit on pump testing.
3. An enlarged diffuser throat, traded against suction margin. The measured throat limited runout and the published optimisation trend [7] both point at the same lever.
4. Hot gas generator products as the working fluid, closing the specific heat ratio gap between cold air at $\gamma = 1.4$ and combustion gas near 1.2, which also shifts the starting margin.
5. A machined metal impeller, to isolate the contribution of the printed surface quality to the measured churning multiplier.
6. A full admission variant at larger diameter, to isolate the partial admission sector losses from the rest of the loss budget.
7. CFD of the as-built geometry with the measured boundary conditions, now that there is experimental data to validate it against, targeting the partial admission windage and the shock boundary layer interaction the 1D tools cannot resolve.
8. Global shutter imaging with a synchronised strobe through the existing transparent casing. The present consumer camera resolves the cavity envelope but not bubble dynamics, since the frame rate sits far below blade passing and the rolling shutter distorts fast structures; individual bubbles already appear in isolated frames, so the optical access is proven and only the camera limits what it can measure.

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9. High rate burst logging on the pressure channels. The logged data are 32 sample averages at 50 Hz, so blade passing unsteadiness at several hundred hertz is unresolvable by design; short bursts at the ADC's native rate would resolve it without changing the architecture.

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A Conventional Impeller Design Attempt

B Supplementary Safety Calculations

The containment and overspeed calculations that sized the protection hardware are in the main body, eq. (59) onwards. This appendix carries the remaining two analyses of the safety case: the venting noise and the venting reaction force. As in the body, the numbers are the 100 bar worst case for the stored air system that was later descoped to 7 bar shop air, retained because the hardware was retained.

B.1 Venting Noise

The sound level of gas venting is estimated per API 521 [46]. The choked mass flow through the vent restriction is

$$\dot{m} = A_{choke}\, p_0 \frac{\Gamma}{\sqrt{T}}, \qquad \Gamma = \sqrt{\frac{\gamma}{R}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{70}$$

and the sound pressure level at 30 m and at distance r follow from

$$L = 5\log_{10}\left(\frac{p_0}{p_e}\right) + 51.5, \qquad L_{30} = L + 10\log_{10}\left(\tfrac{1}{2}\dot{m}a^2\right), \qquad L_p = L_{30} - 20\log_{10}\left(\frac{r}{30}\right) \tag{71}$$

For the worst case 100 bar vent through a 1/8” line this gives 102 dB at 30 m and 131 dB at 1 m. Against the Control of Noise at Work Regulations 2005 limits, nobody is in the room during venting, and the 27 dB SNR ear defenders worn in the control room bring the exposure below the action level.

B.2 Venting Reaction Force

The reaction force of an open vent is

$$F = \dot{m}\, a + A\left(p_{choked} - p_e\right) \tag{72}$$

with a the speed of sound at the choked condition. At 100 bar a 1/2” vent would produce around 1.6 kN, which is why the vent line is reduced to 1/8”, bringing the force to around 100 N against a 50 kg weighted stand, with all vents pointed upward so the reaction acts into the floor.

C Additional Figures and Tables